

WHAT WORKS AND FOR WHOM? EFFECTIVENESS AND EFFICIENCY OF SCHOOL CAPITAL INVESTMENTS ACROSS THE UNITED STATES*

BARBARA BIASI
JULIEN LAFORTUNE
DAVID SCHÖNHOLZER

This article identifies which investments in school facilities help students and which are valued by homeowners. Using novel data on school district bonds, test scores, and house prices across 29 U.S. states and a research design based on narrowly decided elections with staggered timing, we find that increased capital spending in schools significantly improves test scores and is efficient on average. However, the effects vary widely depending on the type of project and the characteristics of the school district. Investments in essential infrastructure, such as HVAC systems or pollutant removal, yield notable improvements in student performance, while expenditures on athletic facilities show no measurable academic benefit. Socioeconomically disadvantaged districts gain disproportionately from capital investments, even after accounting for project type, yet these districts typically underinvest in such projects. *JEL codes:* H41, H75, I22, I24, R30.

I. INTRODUCTION

In the 2019–20 academic year, the United States spent \$90 billion—\$1,760 per student—on the construction and renovation of school facilities. Despite these large sums, facilities differ across districts, and many students attend schools in poor conditions. Several studies have investigated how capital expenditures affect students (as measured by test scores) and homeowners (as

* We thank Jaime Arellano-Bover, Stéphane Bonhomme, Caroline Hoxby, Kirabo Jackson, Karthik Muralidharan, Jesse Rothstein, Kevin Stange, Seth Zimmerman, the editors and four anonymous referees, and audiences at NBER (Education and Public Economics), CEPR (Public Economics Symposium), AEA, AP-PAM, and seminars at various institutions for comments and discussions. We thank Ariel Gelrud, Ariel Hsieh, Chelsea Ilarde, Leon Lufkin, Noa Rosinplotz, Viraj Shukla, and Jessica Xu for outstanding research assistance. We thank Stéphane Lavertu for sharing bond data; and Chuck Amos, CEO of The Amos Group, and Rachel Wisnefski, VP of The Amos Group, for sharing data from schoolboardfinder.org. We are grateful for support from Yale University, The Broad Center at the Yale School of Management, and the Spencer Foundation through Small Grant no. 10016890. All errors are our own.

© The Author(s) 2025. Published by Oxford University Press on behalf of President and Fellows of Harvard College. All rights reserved. For Permissions, please email: journals.permissions@oup.com

The Quarterly Journal of Economics (2025), 1–51. <https://doi.org/10.1093/qje/qja013>. Advance Access publication on February 18, 2025.

measured by house prices) using data from individual districts or states. These studies have reached different conclusions. Some, such as [Neilson and Zimmerman \(2014\)](#), find large positive effects on both student test scores and house prices; others, such as [Cellini, Ferreira, and Rothstein \(2010\)](#), find small effects on house prices and almost no effects on test scores. While a recent meta-analysis has identified positive effects of higher school capital spending on students on average ([Jackson and Mackevicius 2024](#)), there is little evidence on what causes these disparate effects.

There are at least two possible explanations for these disparities. First, schools invest in different types of capital projects. Capital outlays can fund a wide array of projects: fixing a leaking roof and building a football stadium are very distinct types of investments and could produce different effects. Second, capital outlays may benefit certain students more than others. For example, if there are diminishing marginal returns from spending time in safe and comfortable facilities throughout the day, socio-economically disadvantaged students may gain more from attending schools in better conditions ([Rauscher 2020](#); [Enami, Alm, and Aranda 2021](#)).

Understanding what types of projects matter and for whom is crucial for policy makers to decide what projects to prioritize. As [Handel and Hanushek \(2023, 192\)](#) write, it is exactly the variations in these effects that “are central to any interpretation and policy use. Indeed, [they] may provide insights into the mechanisms that could lead to larger impacts . . . by ensuring that funds were used in the most productive way.” Yet a lack of nationwide data and information on fund allocation has so far made it difficult to address this question. As a result, the debate on the effectiveness and efficiency of school capital spending is still wide open.

This article brings new data and evidence to this debate. We study not only whether capital investments matter for students and homeowners, but also what matters and for whom. Our analysis makes use of a particular feature of the funding of capital outlays in the United States: the use of bonds, subject to electoral approval in local referenda and repaid with revenues from local property taxes ([Biasi, Lafortune, and Schönholzer 2021](#)). To estimate the effect of bond authorization, we compile a novel data set with information on school-bond referenda (including the text of the ballot), student test scores, and house prices across 29 states.

We use a research design that exploits variation from close bond elections, while allowing for the presence of repeated elections and dynamic and heterogeneous treatment effects.

We find that, on average, bond authorization raises both test scores and house prices. Yet impacts vary widely across types of projects and districts. Spending on infrastructure renovation or upgrades, such as HVAC systems or roofs, and on the removal of toxic materials from buildings raises test scores but not house prices. Conversely, spending on athletic facilities increases house prices but not test scores. In addition, bond authorization is most beneficial in districts with more disadvantaged student populations, because these districts prioritize bonds that improve learning and are valuable to taxpayers and because they have spent less on capital in the past (hence, they are likely to have facilities in worse condition). However, the benefits for disadvantaged students are larger, even holding spending categories and prior spending amounts constant. These results offer direct guidance to policy makers on how to maximize the impact of school capital investments. They also help reconcile the conflicting findings from previous state-level studies. Replicating these studies, we show that differences in test-score effects across states can be primarily explained by differences in spending items and student characteristics.

Our analysis is made possible by a newly assembled panel data set of school districts. We begin by collecting information on bond referenda from various state offices. These data include the share of votes in favor of the proposal, the proposed investment amount, and the text of each ballot measure that describes the proposed use of the funds. Using textual information from each ballot, we group bonds into eight categories of projects. These include classroom space; infrastructure such as plumbing, roofs, and furnaces; heating, ventilation, and air conditioning (HVAC) systems; IT facilities and labs (STEM); building adjustments to comply with health and safety standards; athletic facilities; purchases of land; and purchases of transportation vehicles, such as school buses.

We link information on bond elections to records from a novel data set with average test scores of students in each district and year. District-level test-score averages, normalized across states, are available starting in 2009 as part of the Stanford Education Data Archive (Reardon et al. 2021). We extend this data set to earlier years, resulting in district-level test-score averages from

as early as 1994 for some states, and for 2003–2019 for nearly all states. We link bond and test-score data to a house price index, constructed by [Contat and Larson \(2024\)](#) at the census tract level and aggregated to the district level, and to enrollment, expenditures, and revenues from the National Center for Education Statistics (NCES). Our analysis data set covers 12,370 bond elections in 29 states and 4,353 districts.

We use these data to estimate the causal effects of bond authorization. To this end, we modify the dynamic regression discontinuity (DRD) design of [Cellini, Ferreira, and Rothstein \(2010\)](#) (CFR) to be robust to heterogeneous effects, for example, due to differences in effectiveness across bond categories or districts. Concretely, we match each district that approved a bond in a given year (or cohort) with all districts that also propose (but do not approve) a measure in the same year. We stack cohorts of treated and control units and estimate the dynamic RD model on this stacked data set, controlling for cohort fixed effects. Our results are robust to a host of alternative estimation methods, including the use of differently defined clean controls, the estimator proposed by [Wooldridge \(2021\)](#), and the original CFR estimator.

Our results indicate that bond authorizations raise capital outlays sharply by about \$1,500 per pupil in the five years after bond authorization. Test scores increase after an authorization, reaching a 0.1 district-level standard deviations higher level after eight years. Two-stage least squares (2SLS) estimates imply that, accounting for the long life of capital projects and a standard rate of depreciation, a \$1,000 per pupil increase in capital spending over five years leads to a 0.04 standard deviation increase in district-level test scores. We also find that a bond authorization leads to changes in the sociodemographic composition of school districts, likely due to household sorting across districts. However, these compositional changes can only account for roughly one-third of the increase in test scores that we document.

These average effects mask important differences in the effectiveness of capital investments across bonds, districts, and students. What the money is spent on plays a crucial role in determining the size of the effects. We obtain category-specific estimates by adapting our stacked DRD design to only compare districts that propose bond measures in the same category. The resulting estimates can only be interpreted as causal under the assumption that the composition of categories of a bond is uncorrelated with the effect size of the category of interest. To support

this assumption, we show that results are similar when controlling for the presence of other categories. These estimates reveal that only some types of expenditures increase test scores: HVAC, safety and health improvements, STEM equipment, infrastructure, and classroom space. For example, authorizing a bond that finances HVAC increases test scores by over 0.20 standard deviations three to six years after the election. The effectiveness of capital spending also depends on who is exposed to the spending increase. The positive effects of bond authorization on test scores are concentrated in districts with a large share of free- and reduced-price lunch eligible students (FRPL, a proxy for low socioeconomic status (SES)) or minority (Black and Hispanic) students. 2SLS estimates confirm that a \$1,000 per pupil increase in spending over five years increases test scores only in more disadvantaged districts. Capital spending is thus most effective in those districts.

Next we examine the effects of bond authorization on house prices to obtain a measure of taxpayer valuation of these investments and to test for spending efficiency. House prices increase by about 9% eight years after an authorization, indicating that homeowners value school capital investments more than the increase in property taxes they are asked to sustain. Assuming no fiscal externalities, this increase would indicate that capital investments are inefficiently low. In our context, it appears to be largely driven by the presence of aid given by states to school districts to supplement local funds. This aid drives a wedge between the amount of money homeowners contribute to the project in the form of property taxes and the actual spending increase, which gets capitalized in the local housing market. In fact, 2SLS estimates on house prices using only locally financed spending as the explanatory variable suggest that homeowners' valuation is similar to this spending increase, which in turn implies that spending levels are on average efficient.

As for test scores, we find large differences in efficiency across spending categories and types of districts. Following a bond authorization, house prices increase for spending on classrooms and athletic facilities but not for other test score-increasing categories. 2SLS estimates confirm that spending on these categories is inefficiently low. This finding implies that learning-enhancing spending does not necessarily increase house prices and investments that capitalize in the housing market do not necessarily enhance learning. This finding contradicts the widespread notion

that housing market effects of school capital outlays are primarily due to academic benefits to students, and it suggests that they may materialize on the basis of other amenities these investments provide to homeowners. Capital spending is also inefficiently low in districts with a large share of FRPL or minority (Black and Hispanic) students.

Together, our results indicate that both the types of projects funded by capital outlays and their beneficiaries shape the impact of increased spending on school facilities. Differences across spending categories, baseline capital stock, and districts' sociodemographic composition are essential in effectively designing and targeting the allocation of funds for school facilities across the United States; ignoring these differences can lead to misguided conclusions about the returns to educational investments. Back-of-the-envelope calculations suggest that closing the spending gap between high-SES and low-SES districts and targeting the additional funds toward HVAC and safety/health (the categories with the highest effects) could reduce the initial gap in test scores among these districts by up to 25%.

This article primarily relates to a literature, spurred by the Coleman report (Coleman et al. 1966), on whether investing more money in schools improves student outcomes. While older studies provide mixed evidence on resource-based policies (e.g., Hanushek 1997; Card and Krueger 1992a, 1992b), more recent research has shown that increasing spending and equalizing spending across districts can improve educational outcomes (e.g., Jackson, Johnson, and Persico 2016; Hyman 2017; Lafortune, Rothstein, and Schanzenbach 2018; Candelaria and Shores 2019; Jackson 2020), labor market outcomes (Jackson, Johnson, and Persico 2016), and intergenerational mobility (Biasi 2023). As we do, some of these studies have used variation from close elections to identify the effects of increased current and operational spending (Abott et al. 2020; Baron, Hyman, and Vasquez forthcoming). We contribute to this literature by showing that, across most U.S. states, increased spending on capital projects improves student outcomes and is valued by homeowners. We also demonstrate empirically that properly accounting for the use of funds and the characteristics of the students who experience the funding increase is crucial to establishing whether and how money matters.

Our work is also related to a set of studies, pioneered by Cellini, Ferreira, and Rothstein (2010), that have estimated the effects of school capital expenditures on students and the real

estate market, reaching conflicting conclusions. Most of these studies use data from individual states (Cellini, Ferreira, and Rothstein 2010; Goncalves 2015; Hong and Zimmer 2016; Conlin and Thompson 2017; Rauscher 2020; Enami, Alm, and Aranda 2021; Baron, Hyman, and Vasquez forthcoming) or school districts (Neilson and Zimmerman 2014; Lafortune and Schönholzer 2022). The only exception is Brunner, Hoen, and Hyman (2022), who use variation in revenues from wind energy installations to study test-score effects across states. Most of these studies (including Brunner, Hoen, and Hyman 2022) find small (and often imprecise) effects of capital spending, whereas Neilson and Zimmerman (2014) and Lafortune and Schönholzer (2022) find larger and positive effects. This article reconciles this literature by showing that capital spending can have profoundly different impacts depending on the types of projects and the characteristics of the students who benefit from them. More broadly, our results also speak to a literature that has estimated the capitalization of school quality into house prices (see Black and Machin 2011, for a review).

Last and most important, this article brings empirical evidence to the recent debate over the drivers of the observed differences in the impacts of school spending, particularly evident in the context of capital outlays (Handel and Hanushek 2023; Jackson and Mackevicius 2024). Since the precise use of capital funds is generally not recorded in district administrative data, prior studies have been unable to distinguish the effects of different spending items beyond operational and capital spending. Using newly collected data on the text of ballots from bond referenda, we can classify spending into much finer categories. With these data, we uncover large differences in effects across categories and across districts serving different populations of students. Our findings can be used by state and district officials to understand how to best target school capital investments, with the ultimate goal of maximizing the returns for students and taxpayers.

II. SCHOOL CAPITAL EXPENDITURE ACROSS THE UNITED STATES

U.S. public school districts allocate about 10% of their budget each year to capital outlays (Cornman et al. 2021). Funding for capital projects is governed by specific rules that set it apart from other school expenditures. Roughly three-fourths of capital

outlays are funded locally (Filardo 2016), compared to only 45% of current spending (U.S. Department of Education 2023). In addition, while state-level school finance reforms have led to a more equal and progressive distribution of current expenditures across school districts, the distribution of capital outlays has remained unequal. Spending on capital outlays varies substantially across and within states (Online Appendix Figure A1). The precise rules school districts must follow to raise funds for capital projects differ across states.¹ We summarize them here.

II.A. Local Bond Elections

In most U.S. states, capital outlays are primarily funded using bonds issued by each school district. In every state except Hawaii, Kentucky, and Massachusetts, bonds must be approved by voters in local referenda, usually (but not always) held during a primary or general election.² States differ in the majority required for a bond proposal to pass. Thirty-six states require a simple majority, that is, 50% of those who turn out to vote. The remaining 11 states require a supermajority, ranging from 55% in California to 67% in Idaho (Online Appendix Figure A2). With the exception of New Hampshire (which reduced its required majority from two-thirds to 60% in 1999) and California (which reduced it from two-thirds to 55% in 2001), required majorities did not change between 1995 and 2017, the time period we study.

In local bond referenda, ballots outline the proposed use of the funds, typically reflecting the greatest capital needs of the district. Ballots also summarize project costs and mention the projected increase in local property taxes. For example, voters in the Templeton Unified School District, CA were called to vote on the following bond proposal in 2012:

To update classrooms with modern computers/technology; replace portable classrooms; expand vocational education facilities including welding, engineering, medical/science technology, and construction trades; and renovate, upgrade, equip, acquire and construct classrooms/school facilities throughout the District; shall Templeton Unified School District be authorized to issue \$35,000,000 of

1. See Biasi, Lafortune, and Schönholzer (2021) and Blagg, Terrones, and Nelson (2023) for a summary of these rules.

2. The exact election timing is a decision of each school district. In some states, districts can also call referenda to increase local property tax rates to fund operational expenses. We do not consider these elections in our analysis.

bonds with interest rates below legal limits, an independent citizen oversight committee, no money for administrator salaries, and all funds spent locally and not taken by the State and spent elsewhere?

In this referendum, 57.8% of voters approved the measure. Since California has a required majority of 55%, the proposal was authorized and the district was able to issue bonds. Over the next several years, the district's schools were modernized with these funds.

Districts may propose and pass several bond measures over time. Districts who fail to approve a measure may also choose to hold another election shortly thereafter.³

II.B. State Aid

State aid represents less than 30% of funds for capital projects on average, and less than 5% in about half of the states. Yet states such as Alaska, Hawaii, Maine, Massachusetts, New Hampshire, New York, Rhode Island, and Wyoming primarily fund capital with state dollars.

States that support school capital projects typically do so with grants to school districts. Twenty-seven states use conditional grants, contingent on a district's ability to raise funds locally. For example, California funds 50% to 60% of the cost of eligible capital projects to districts that raise local funds. Other states use unconditional grants, funded through sales taxes, state bond revenues, and general-fund appropriations. While conditional grants tend to be regressive, some unconditional ones are progressive and allocate larger sums to districts with lower property tax bases. Ohio, for example, distributes funds to districts based on local property wealth and household income.

III. DATA

Our analysis uses a new panel data set of U.S. school districts with information on bond elections, school district finances,

3. In general, districts that succeed in authorizing a measure may choose to either fully exhaust their bonding capacity up to the limit approved by the voters or to do so gradually. Districts may also choose to propose several small bonds in short succession (rather than a single large bond) to fund a project. The Los Angeles Unified school district adopted this strategy in the late 1990s and early 2000s, passing several bonds to fully fund a \$25 billion, multidecade infrastructure renewal project (Lafortune and Schönholzer 2022).

enrollment, and demographics, test scores, and house prices. We link it to the funding rules in place in each district and year, summarized in [Biasi, Lafortune, and Schönholzer \(2021\)](#). We describe each set of variables and their respective sources. We refer to each academic year with the calendar year of the spring semester (e.g., 2017 for 2016–17).

III.A. School-Bond Elections

We created a unified nationwide database of school-bond referenda by combining information from various sources. In most states, records of the most recent local elections are published on the websites of agencies and offices such as the Secretary of State, the Department of Education, and the Department of Elections. We collected online records and complemented them with those obtained through formal public data requests. We further added data on bond elections for nine states from [Abott et al. \(2020\)](#).

The resulting database covers elections in 41 states. Most records include the date of the referendum, the share of votes in favor of the measure, the proposed bond amount, and the ballot text. We discard data from eight states because they are incomplete and from three additional states because they fail to pass statistical tests required for the validity of our empirical strategy. We also exclude Massachusetts because it does not have mandatory voting on bonds. The availability of data, their sources, and details on the construction of our final sample are described in [Online Appendix Table B1](#).

To match the rest of our data, we aggregate election information at the district-year level; if a district has multiple bonds in a year, we consider either the largest bond (for bonds with information on the spending amount) or the bond with the vote share closest to the majority (for bonds with no information on the amount).⁴ The resulting data set is an unbalanced panel of 10,146 districts in 29 states, enrolling about 71% of all U.S. students, described in [Table I](#), first column. The panel begins as early as 1990 for some states and 2003 for all states, and it ends in 2017.⁵ In these data 4,683 districts (46%) propose at least one bond measure over the sample window (second column); our stacked approach (which we describe later) uses information from 4,353 of

4. This approach follows [Martorell, Stange, and McFarlin \(2016\)](#).

5. [Online Appendix Figure A4](#) shows the number of states with district bond data and the number of bonds in each year.

TABLE I
DISTRICT EXPENDITURES, BONDS, AND SPENDING CATEGORIES: SUMMARY
STATISTICS

	Full sample	W/ election	Analysis sample
Capital	1,320.0 (2,895.2)	1,407.1 (3,051.2)	1,614.0 (3,517.2)
Current	7,046.9 (3,940.0)	6,560.1 (2,597.6)	7,003.8 (3,024.4)
Spending rules			
Share w/ supermajority	0.19 (0.39)	0.22 (0.42)	0.22 (0.41)
Voting requirement	0.51 (0.034)	0.52 (0.034)	0.52 (0.033)
Debt limit	0.095 (0.057)	0.096 (0.058)	0.095 (0.058)
Bonds			
Share proposing a bond/year	0.062 (0.24)	0.11 (0.32)	0.19 (0.39)
Share approved	0.75 (0.43)	0.75 (0.43)	0.77 (0.42)
Vote margin	0.099 (0.16)	0.099 (0.16)	0.11 (0.16)
Size per pupil proposed (\$1,000)	7.67 (8.26)	7.54 (8.14)	7.66 (8.48)
Categories, approved bonds			
Classrooms	0.45 (0.50)	0.45 (0.50)	0.45 (0.50)
Other infrastructure	0.27 (0.44)	0.27 (0.44)	0.27 (0.44)
HVAC	0.12 (0.32)	0.12 (0.32)	0.12 (0.32)
STEM equipment	0.28 (0.45)	0.28 (0.45)	0.28 (0.45)
Safety/health	0.20 (0.40)	0.20 (0.40)	0.21 (0.41)
Athletic facilities	0.17 (0.38)	0.17 (0.38)	0.18 (0.38)
Transportation	0.31 (0.46)	0.31 (0.46)	0.34 (0.47)
Land purchases	0.13 (0.33)	0.13 (0.33)	0.13 (0.33)
Demographics and outcomes			
Share low-SES	0.39 (0.22)	0.38 (0.22)	0.42 (0.22)
Share Black/Hispanic	0.22 (0.26)	0.23 (0.26)	0.26 (0.27)
ELA test scores	-0.055 (0.92)	-0.067 (0.91)	-0.080 (0.92)
Math test scores	-0.12 (0.92)	-0.14 (0.92)	-0.13 (0.93)

TABLE I
CONTINUED

	Full sample	W/ election	Analysis sample
House price index (1989 = 100)	168.9 (57.5)	175.4 (61.2)	196.1 (60.0)
Number of districts	10,146	4,683	4,353
Number of states	29	29	29

Note. Means and standard deviations (in parentheses) of variables of interest. In all columns, the sample is restricted to states that pass the McCrary test.

these districts (third column). Each year, 19% of these districts propose a measure, and 77% of these proposals are authorized (Table I). Bonds are comparable in terms of interest rates, with a standard deviation of 0.66% (Online Appendix Figure A3).

1. *Classifying Bonds into Categories.* We assign each bond to one or more spending categories using information from the ballot text. To obtain a list of categories that is both informative and contained, we rely on the classification produced by the website [SchoolBondFinder.com](https://www.schoolbondfinder.com) (SBF), created and managed by the Amos Group, which collects detailed information on approved bonds since 2014. SBF groups bonds into six categories (construction and renovation, capital improvements, safety and health, technology, transportation, and others) using a proprietary algorithm. We modify this list by (i) dividing capital improvements into investments in HVAC (shown by [Park et al. 2020](#) to be crucial for student learning) and investments in other infrastructure (plumbing, roofing, and furnaces); (ii) dividing facility constructions and renovations into those pertaining to classroom space and those pertaining to athletic facilities; and (iii) adding land purchases as a separate category. Excluding “others,” this leaves us with eight categories: classroom construction and renovation (“classroom” in short); HVAC; other infrastructure; safety and health; technology, IT, and laboratory spaces (or STEM); athletic facilities; land purchases; and transportation.

To classify bonds into categories, we perform a simple search of the keywords listed in [Online Appendix B.2](#) in the text of each ballot. Manual checks on a random sample of 200 ballots indicate that this procedure does an excellent job in allocating bonds to categories, with an error rate of less than 1%. We successfully assign 75% of the 14,000 bonds in our final data to at least one

category based on the presence of each category's keywords in the ballot text (details in [Online Appendix B.2.1](#)).⁶ It is important to notice that our category indicators are not mutually exclusive, so each bond could be linked to more than one category.⁷

III.B. District Finances, Enrollment, and Demographic Information

Data on district finances are from the Annual Survey of School Districts of the NCES and from the Census of Governments. We use information on districts' total expenditure, expenditure by category (capital, current instructional, and current noninstructional), and revenues by source starting from 1995, measured in 2020 US\$ per pupil. We also use demographic information for each school district and year from the NCES Common Core of Data (CCD). These include enrollment, racial and ethnic composition of the student body, and share of low-income students (defined as those eligible for FRPL).

III.C. Student Achievement

Our analysis requires measures of student achievement in all districts across all years. In the United States, all students in grades 3 to 8 take state standardized exams in math and either reading or English language arts (ELA). These exams differ across states and years, making comparisons along these dimensions very challenging. A notable exception is the NAEP, taken annually by a sample of grade 4 and 8 students in a subset of all districts. By design, NAEP scores are not available for all districts.

In an attempt to create a comprehensive district-level panel of student achievement, [Reardon et al. \(2017\)](#) and [Fahle et al. \(2021\)](#) converted state test scores into a uniform national scale, normalizing them across states and years using moments from the distribution of the NAEP. They applied this method to data from standardized state exams for 2009–2018. The resulting normalized scores are publicly available as part of the Stanford Education Data Archive (SEDA).

6. The majority of unassigned bonds have uninformative or incomplete ballot texts, mostly due to apparent recording mistakes.

7. [Online Appendix Table A1](#) shows the distribution of the number of categories, across all bonds and separately for bonds containing the eight categories we consider.

We extend this panel backward by adding data from various sources. For 2001–2005, we use data from the National Longitudinal School-Level State Assessment Score Database (NLSLSASD), originally maintained by the U.S. Department of Education and now discontinued. The NLSLSASD contains average test scores at the school-subject-grade-year level for nearly every state.⁸ For 2006–2008, we collected test-score averages from each state’s education department either via direct download from the departments’ websites or via public data requests. These data are available at the district or at the school level, separately by grade, subject, and year. We aggregated data from NLSLSASD and state departments at the district-subject-grade-year level and normalized it using the same procedure as SEDA, described in [Online Appendix B.1.1](#).

By combining data from these three sources, we build a panel data set of standardized average test scores in math and reading/ELA for grades 3–8 for 10,146 districts with bond election data, going back to 2003 for all states and as early as 1994 for some. Because test scores are standardized, all estimates are expressed in district-level standard deviations.⁹ On average, these districts enroll 71% of all students in the nation.

III.D. House Prices

We measure changes in the real estate market with a house price index (HPI), constructed by [Contat and Larson \(2024\)](#) using a repeat-sales approach applied to data from Fannie Mae and Freddie Mac, the Federal Housing Administration, and county recorder rolls provided by CoreLogic. This HPI is available for a balanced panel of 63,122 census tracts for 1989–2021.¹⁰ It is normalized to a value of 100 in 1989 for each tract and grows according to repeat-sales estimates in the tract or nearby tracts. To aggregate the data to the school district level, we map census tract centroids to 2010 school district boundaries from the

8. See [Online Appendix Figure A5](#) for a map of the first available year of data for each state.

9. On average, school and district-level standard deviations are smaller than student-level standard deviations ([Kraft 2020](#)). Using a subset of our data, we estimate that student-level standard deviations are on average about 2.86 times larger than district-level ones; accordingly, our estimates would be roughly 2.86 times smaller if they were estimated using student-level data (though this scale factor varies across states, grades, subjects, and years).

10. These tracts are based on the 2010 census tract geography.

NCES Education Demographic and Geographic Estimates Program (EDGE) and calculate the average house price index for each school district and year. This procedure yields a balanced panel of 4,679 school districts with bond-election information for the period 1990–2017, enrolling 61% of all students.

IV. ESTIMATING HETEROGENEOUS CAUSAL EFFECTS OF BOND AUTHORIZATION

Our goal is to estimate the causal effect of bond authorization, allowing it to be dynamic over time and differ across bonds. We set up a research design that allows us to achieve this goal. Our starting point is a simple comparison of outcomes over time between “treated” districts (i.e., those that succeed in authorizing a bond), and “control” districts that propose a bond in the same year but fail to authorize it.

There are three challenges in pursuing this goal. First, treated and control districts may differ in ways that are unobservable to us. Second, districts may propose and pass multiple bonds in our time period of analysis, and the likelihood of proposing a new bond may be related to both the success and the impacts of previous proposals. Third, because districts that fail to pass a bond in a given year might successfully do so in the future, comparing treated and control districts may result in “forbidden comparisons” (Borusyak, Jaravel, and Spiess 2024), which have been shown to yield biased estimates if treatment effects are correlated with treatment timing. The first two challenges have already been recognized and addressed in the literature, starting with Cellini, Ferreira, and Rothstein (2010). The third, particularly relevant for our research question, has not. We therefore review solutions to the first two challenges and propose a solution to the third.

IV.A. Addressing Challenge 1: Exploiting Variation from Close Elections

We begin by considering the simple case of districts that propose and authorize at most one bond measure over the period of study. Let V_{jt} be the share of votes in favor of a bond measure proposed by district j in year t , v the required share of favorable votes to authorize the measure, and $D_{jt} \equiv \mathbb{1}(V_{jt} \geq v)$ an indicator for bond authorization in t . The effect of bond authorization on

an outcome Y_{jt} measured k years after the authorization, denoted as β_k , is the difference in outcomes between treated districts and their counterfactual had they failed to pass a bond in that year. A standard difference-in-difference setup around the timing of bond authorization would then have the form

$$(1) \quad Y_{jt} = \alpha_j + \gamma_{s(j)t} + \sum_{k \neq 0} \beta_k D_{jt-k} + u_{jt},$$

where α_j and $\gamma_{s(j)t}$ are district- and state-by-year fixed effects and u_{jt} is an error term. We normalize β_0 to be zero. If $\mathbb{E}(u_{jt}|D_{jz}) = 0$ (in other words, if outcomes of treated and control units would have been on similar trends in the absence of the treatment), OLS estimates of β_k capture the effects of bond authorization.

This assumption is unlikely to hold in our context. Districts that succeed in authorizing a bond may differ from those that fail in many unobservable ways. For example, they may be more successful in securing funds for capital expenditures through state grants; they may have a different history of capital investments; or they may serve a different body of students. All these differences could place treated and control districts on different trajectories, violating the parallel-trends assumption of difference-in-differences (i.e., $\mathbb{E}(u_{jt}|D_{jz}) \neq 0$). OLS estimates of β_k in [equation \(1\)](#) would then be biased.

To overcome this challenge, CFR propose a dynamic RD framework that exploits close elections. The intuition is as follows. Since the probability of authorizing a bond jumps discontinuously at the cutoff v , if unobserved characteristics (captured by $\mathbb{E}(u_{jt}|V_{jz})$) are continuous around v , then districts that fail to authorize a bond proposal by a thin margin are a good counterfactual for districts that succeed by a thin margin. Under this milder assumption, we can consistently estimate β_k via OLS on an augmented version of [equation \(1\)](#), which controls for polynomials of the vote margin in the years before and after t . We denote these polynomials with $P^g(V_{jt-k}, \delta_k^g)$, where g is the order of the polynomial and δ_k^g are its coefficients. As CFR argue, this framework identifies the effects of bond authorization from close elections.¹¹

11. Close elections tend to occur in districts that spend less on average, in states with a supermajority requirement, and for larger bonds ([Online Appendix Table A2](#)). Following CFR, to improve power we retain all data in estimation, rather than focusing on observations around the cutoff v .

IV.B. Addressing Challenge 2: Controlling for Bond Histories

In reality, matters are more complicated than this simple case because districts can propose and authorize multiple bond measures over time. On average, in our sample period 66% of all districts that propose at least one measure do so more than once; among these, 49% authorize more than one measure. As a result, both treated and control districts may have been treated in the past or may become treated in the future. In this case, estimates of [equation \(1\)](#) capture the direct effect of bond passage and the indirect effect of past and future bonds. Proposals and authorizations may also be correlated over time. In our data, districts that fail to authorize a bond at time t on average propose a new bond after 2.9 years and authorize one after 3.1 years. Districts that authorize a bond propose and authorize new bonds 3.9 and 4.0 years later, respectively ([Online Appendix Figure A6](#)).

CFR extensively discuss this issue and propose a solution: a treatment-on-the-treated (TOT) estimator that captures the effect of bond authorization against the counterfactual of never authorizing a bond in the foreseeable future. The TOT corresponds to the OLS estimator of β_k in a version of [equation \(1\)](#) that includes controls for a district's history of bond proposals:

$$(2) \quad Y_{jt} = \alpha_j + \gamma_{s(j)t} + \sum_{k \neq 0} [\beta_k D_{jt-k} + P^g(V_{jt-k}, \delta_k^g) + \phi_k M_{jt-k}] + u_{jt},$$

where M_{jt-k} equals one if the district proposed a bond measure k years before t . In practice, controlling for M_{jt-k} ensures that we estimate treatment effects by comparing districts with the same bond history.¹² This estimator is consistent if effects are (i) additive across bonds, and (ii) heterogeneous across time elapsed since the election but homogeneous across bonds.¹³ The latter assumption is required because the TOT uses late-treated districts

12. The term M_{jt-k} for $k < 0$ may seem like a “bad control” (i.e., a control that is also affected by the treatment) if future bond elections are affected by the outcome of the focal election. Not including these terms implies that our RD estimates would capture the effects of authorizing a bond in the focal year and in future years, which is not what we want to measure. Reassuringly, whether or not we include variables for future bond elections has little effect on our estimates (see [Online Appendix Figures A21, A22, A29, and A33](#)).

13. CFR state that “We assume (as has been implicit in our notation thus far) that the TOT effects of bond authorization on later authorizations and outcomes depend only on the time elapsed since the focal treatment (τ) and not on the time at which the treatment occurred or on the treatment history” (229).

as controls for early-treated districts. Early-treated districts may continue to experience the effects of bond passage by the time late-treated districts pass a bond. If treatment effects vary depending on the timing of the election, these long-run effects are not properly accounted for, and estimates could be biased.

IV.C. Addressing Challenge 3: Stacked DRD

Assuming homogeneous treatment effects is not ideal in our context. For example, this assumption could be violated if bonds' effects vary across spending categories. To relax it, we build on a recent literature (including [Callaway and Sant'Anna 2021](#); [Goodman-Bacon 2021](#); [Sun and Abraham 2021](#); [Wooldridge \(2021\)](#); [Borusyak, Jaravel, and Spiess 2024](#)) that has dealt with estimating treatment effects when treatment timing is staggered and effects are correlated with the timing of the treatment. The intuition is to compare units that become treated in t with units that are never treated after t , avoiding comparisons with units that become treated at a later time. In our context, this requires comparing districts that (barely) authorize a bond in a given year (or treatment cohort) with districts that also propose a bond in the same cohort but fail to authorize it and never authorize any bonds in the future (either because they do not propose any bonds or because they propose but fail to pass them). We implement this strategy with the following steps.

- (i) For each treatment cohort c , we select treated districts and untreated ones that (a) also propose a measure in c but do not authorize it, and (b) do not authorize any bond in the 10 years after an election (regardless of whether they propose any). For all these districts, we only retain data for the 5 years preceding c and the 10 years following it.
- (ii) We “stack” data for each cohort to form a larger data set, where one observation corresponds to a district, year, and cohort.¹⁴ We then estimate the following equation via OLS on the resulting data set:

$$Y_{jct} = \alpha_{jc} + \gamma_{s(j)ct} + \sum_{k \neq 0} [\beta_k D_{jct-k} + \phi_k M_{jct-k} + P^g(V_{jct-k}, \delta_k^g)] + u_{jct}. \quad (3)$$

14. This “stacked” approach has become popular in the context of difference-in-differences with staggered timing. Two of the earliest examples are [Cengiz et al. \(2019\)](#) and [Deshpande and Li \(2019\)](#).

This strategy addresses all three challenges. First, including leads and lags of polynomials of the vote margin implies that the parameters β_k are identified and estimated using quasi-random variation from close elections. Second, the inclusion of M_{jct-k} holds fixed any factors that may induce districts to propose a measure at a given point in time. Last, the comparison of treated districts with clean controls (which avoids forbidden comparisons) ensures that our estimates are consistent even when treatment effects are heterogeneous across units (Dube et al. 2023). In the estimation, we weight observations by the number of test score takers (for test scores) or district enrollment (for capital spending or house prices) in the first year of the sample.¹⁵

Two details of this strategy are worth emphasizing. First, our definition of clean controls as units that remain untreated over time does not introduce endogenous selection in the sample. Clean controls are districts that propose bonds in the same years as the treated districts (due to the inclusion of leads and lags of M_{jct-k}), but *barely fail* to approve them for reasons that are as good as random (because of the inclusion of leads and lags of $P^g(V_{jct-k}, \delta_k^g)$, which helps isolate variation from close elections in the focal year, in the past, and in the future). To further ensure that there is no selection on unobservables in the group of districts that barely fail in c and continue to propose a bond in the future (those that share a bond history with treated districts), we estimate a version of our model where we do not control for M_{jct-k} for $k < 0$ (i.e., we do not control for a district's future bond history).¹⁶

15. This weighting scheme allows us to get at the average effect of bond approval on each student. Because our data are aggregated at the district level, without enrollment weights we would give disproportionately less weight to students in very large school districts (such as Los Angeles, CA, enrolling more than 600,000 students) compared with those in small districts (such as Gorman, CA, enrolling fewer than 100 students). In addition, district-level test-score averages and the house price index are noisy estimates of true parameters. By weighting by enrollment, we down-weight less precise estimates (i.e., those obtained from smaller samples). Estimates are robust to not using any weights.

16. This could occur, for example, if districts that barely fail to authorize a bond and continue to propose one have larger potential effects from bond authorization or if authorizing a bond changes the electorate in a way that affects the presence and outcome of future elections. Online Appendix Figure A10 shows that the characteristics of the electorate remain fairly stable following a bond authorization.

Second, our strategy considers all the elections of a given district as separate events and does not restrict the number of authorizations for treated units in each relevant time window (in other words, the sum of D_{jt-k} across all k need not sum to one for each observation in the data set). We thus maintain the assumption of additive separability of the effects of subsequent authorizations.

1. *Alternative Designs.* We probe the robustness of our results to five alternative research designs. First, we implement the TOT estimator of CFR. Second, we estimate a version of our stacked DRD design where we define clean controls as districts that do not approve any bonds in the entire $[c - 5, c + 10]$ time window. Third, we estimate a version of our stacked DRD design where we define clean controls as districts that have the same bond history as at least one treated district. Fourth, we estimate our stacked DRD not controlling for future bond history (i.e., excluding M_{jct-k} for $k < 0$ from equation (3)). Last, we implement the extended two-way fixed effects (ETWFE) estimator proposed by Wooldridge (2021), which allows β_k in equation (2) to vary by cohort. Our results are largely robust to the use of these approaches.¹⁷

IV.D. Testing the Validity of the Research Design

Our empirical strategy requires electoral outcomes to be as good as randomly distributed among districts with close elections. We examine the plausibility of this assumption in three ways. First, we perform a McCrary (2008) test of smoothness of the density of the vote share around the electoral cutoff. A discontinuity could indicate endogenous sorting around the cutoff, which would violate the RD assumption. State-specific histograms of the vote margin (the difference between the vote share and the required majority) show discontinuities at zero in Arkansas, Missouri, and Oklahoma but not in other states (Online Appendix Figure A7).¹⁸

17. The only exception are effects on house prices estimated as in CFR. This difference is likely due to the fact that, as the authors argue, their estimator is not robust to the presence of heterogeneity in treatment effects across election cohorts, whereas the other estimators are.

18. These discontinuities appear to be driven by a lack of mass to the left of the cutoff. This data pattern could be attributable to the unavailability of election data for failed referenda. Alternatively, it could occur if all referenda pass in these states.

We exclude these states from our analysis. The resulting pooled density function is smooth around zero both in the main data and in the stacked data used for estimation (Figure I, Panels A and B, respectively).

Second, we test for the smoothness of pre-election district covariates around the cutoff. We consider average household income, the share of people with at least a college degree, the shares of FRPL and white students in the district, enrollment in private schools, total revenues, state revenues, and total expenditure. All these variables are smooth around the cutoff in the main and in the stacked data set (Online Appendix Figures A8 and A9).

Third, we examine pre-election differences in outcomes between treated units and their clean controls, captured by β_k for $k < 0$ in equation (3). The absence of significant differences (which we show and discuss later) suggests that trends in clean controls are a good counterfactual for treated units. Taken together, these tests support the assumption of quasi-random assignment to treatment among districts with close elections.

V. EFFECTIVENESS OF SCHOOL CAPITAL SPENDING

We begin by estimating the effectiveness of bond authorization for student learning. We present effects on capital spending per pupil (the first stage); then we turn to our main measure of effectiveness: test scores. We briefly discuss the consequences of bond authorization on interdistrict household sorting.

V.A. First Stage: Effects on Capital Spending

Capital expenditures increase sharply after a bond authorization. Estimates of β_k in equation (3), using per pupil capital spending as the dependent variable, indicate that the difference in spending between districts that barely authorize and those that barely reject a bond measure in year c is on a relatively flat trend in the three years before an election. It increases sharply by \$600 a year at $c + 2$ and by \$590 at $c + 3$, returning to pre-election levels five years after the election (Figure II, line with circle markers).¹⁹

19. Six to eight years after an election, spending is actually lower in treated districts (although this difference is indistinguishable from zero; Table II, column

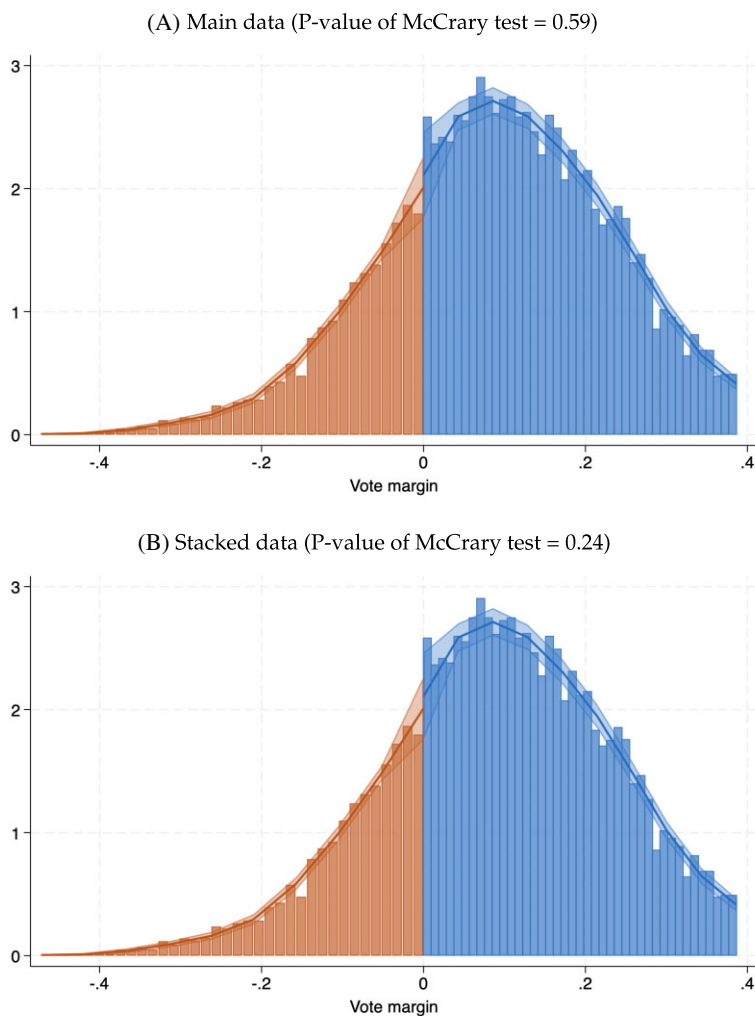


FIGURE I

Smoothness of the Density Function of the Vote Margin

Histograms of the vote margin, defined as the difference between the share of votes in favor of the proposed measure and the required majority in each state. The lines and confidence intervals visually show the result of a [McCrary \(2008\)](#) test for the discontinuity in the density function at zero, using a uniform kernel and a cubic polynomial. Panel A is constructed using the main data set. Panel B is constructed using the stacked data set used in estimation. The sample includes AZ, CA, CO, CT, DE, FL, GA, ID, IN, IA, LA, MD, MI, MN, MS, NC, NE, NV, NY, ND, OH, OR, PA, RD, TX, VA, WA, WV, WI.

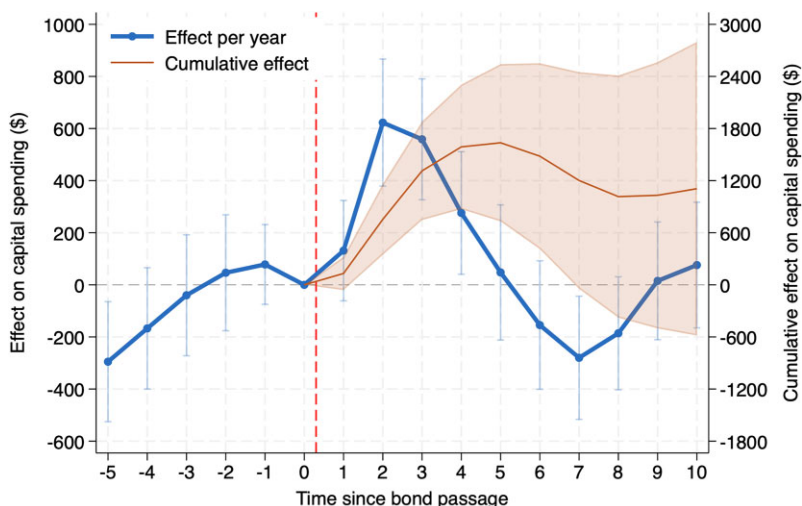


FIGURE II

Average Effects of Bond Authorization on Capital Spending

The blue line with circle markers shows estimates and confidence intervals of the parameters β_k in [equation \(3\)](#), obtained using capital spending per pupil as the dependent variable. The orange continuous line shows cumulative effects, calculated as the running sum of coefficients since time 0. Estimates are obtained using district-by-cohort and cohort-by-state-by-year effects; observations are weighted by district enrollment. Standard errors are clustered at the district level.

Five years postelection, cumulative spending is \$1,650 higher in districts that authorized a bond compared with the years before the election ([Figure II](#), continuous line).²⁰

(1)). A possible explanation for this finding is that control districts manage to increase their capital spending by drawing money from alternative sources to compensate for the lost bond revenues.

20. In line with the presence of small state aid, state revenues earmarked for capital projects also increase after a bond authorization ([Online Appendix Figure A12](#)).

TABLE II
FIRST STAGE: EFFECTS OF BOND AUTHORIZATION ON SCHOOL EXPENDITURES

Type of expenditure:	Capital (1)	Current (2)	Other noninstructional services (3)
Average effect over:			
1–5 years	325*** (93)	7 (30)	3 (4)
6–8 years	–239** (104)	–32 (47)	7 (7)
9–10 years	2 (110)	–64 (51)	10 (8)
District-cohort fixed effects	X	X	X
Year-state-cohort fixed effects	X	X	X
Mean of dep. var.	1,617	7,787	484
Adj. R^2	0.293	0.968	0.863
N	141,341	141,341	141,341

Notes. Estimates and standard errors of linear combinations of the parameters β_k in equation (3). The dependent variables are per pupil capital spending (column (1)), current spending (column (2)), and spending on noninstructional services (column (3)). All columns control for district-by-cohort and cohort-by-state-by-year effects. Observations are weighted by district enrollment. Standard errors in parentheses are clustered at the district level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Bond authorization does not affect noncapital (i.e., current) spending (Table II, column (2) and Online Appendix Figure A11).²¹ This result is unsurprising, as revenues from capital bonds may typically not be used to fund current spending. We can safely interpret the impact of bond authorization on test scores as the effect of increased capital spending. We present these effects next.

V.B. Mean Effects on Student Achievement

To estimate the impact of bond authorization on test scores, we pool data from math and ELA tests taken in grades 3–8. We adapt equation (3) to accommodate this feature of the data:

$$Y_{jrwt} = \alpha_{jc} + \gamma_{s(j)rwt} + \sum_{k \neq 0} [\beta_k D_{jct-k} + \phi_k M_{jct-k} + P^g(V_{jct-k}, \delta_k^g)] + u_{jrwt}, \quad (4)$$

21. Online Appendix Figure A11 shows estimates of β_k in equation (3) using current spending (Panel A) and instructional spending (Panel B) as the dependent variables.

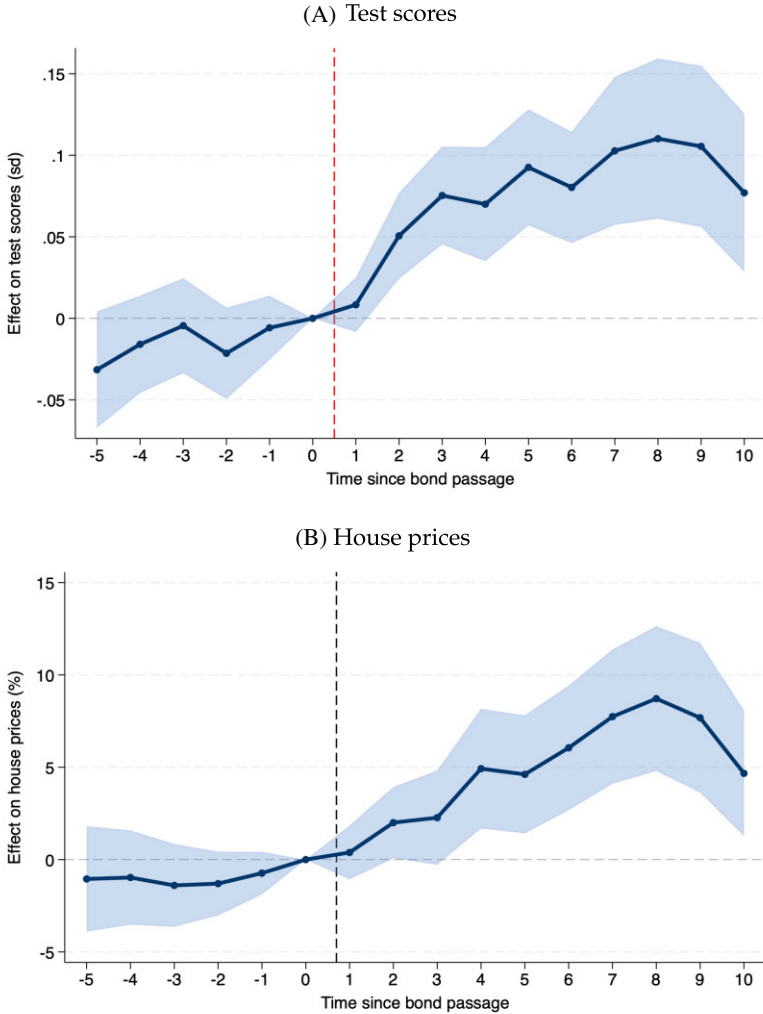


FIGURE III

Average Effects of Bond Authorization on Test Scores and House Prices

Estimates and confidence intervals of the parameters β_k in equation (3), obtained using test scores (Panel A) and the house price index (Panel B) as the dependent variable. Test-score estimates are obtained pooling data across subjects and grades, controlling for district-by-cohort and cohort-by-state-by-year-by-subject-by-grade effects, and weighing observations by the number of test takers. House price estimates are obtained using district-by-cohort and cohort-by-state-by-year effects, weighing observations by district enrollment. Standard errors are clustered at the district level.

where Y_{jrwct} is the standardized average student test score of district j and cohort c for all students in grade r , subject w , and year t . $\gamma_{s(j)rwct}$ contains state-by-grade-by-subject-by-cohort-by-year fixed effects. Everything else in the equation is as before. We weight observations by the number of test-takers and cluster standard errors at the district level.

Estimates of β_k show a significant improvement in student achievement after a bond authorization. The difference in test scores between districts that marginally approve and those that marginally reject a bond measure is constant in the five years leading to an election. It then starts to increase two years after the election, reaching a 0.1 standard deviations higher level eight years postelection (Figure III, Panel A).²²

Table III summarizes the effect of bond authorization on test scores. In districts that marginally approve a bond proposal, test scores are 0.05 standard deviations higher on average 1–4 years after an election, 0.10 standard deviations higher 5–8 years after, and 0.09 standard deviations higher 9–12 years after, relative to districts that marginally reject it (Table III, column (1)). The impact of bond passage is slightly higher for ELA (a 0.12 increase 9–12 years postelection) compared with math (a 0.07 increase 9–12 years after an election), Table III, columns (2) and (3)).

1. *Effect of a \$1,000 per Pupil Increase in Capital Spending on Test Scores: 2SLS.* The literature on the impact of school resources has focused on changes in outcomes per dollar of increased spending as a policy-relevant parameter (see Jackson and Mackevicius 2024 for a review). We transform our reduced-form stacked DRD estimates into a per dollar impact using a 2SLS model. In the first stage, we use the stacked DRD of equation (4) to predict per pupil capital spending in each district, cohort, and year K_{jct} . In the second stage we express test scores in year t as a function of cumulative spending per pupil over the previous

22. These results could likely understate the true effects of capital spending, since only a subset of students in each district may be affected by the investments in practice.

TABLE III
EFFECTS OF BOND AUTHORIZATION ON STUDENT ACHIEVEMENT AND HOUSE PRICES

	Test scores			HPI (4)
	Pooled (1)	Math (2)	ELA (3)	
Average effect over:				
1–4 years	0.051*** (0.014)	0.054*** (0.018)	0.050*** (0.014)	2.397* (1.287)
5–8 years	0.096*** (0.023)	0.096*** (0.028)	0.102*** (0.022)	6.783*** (2.047)
9–12 years	0.091*** (0.029)	0.068** (0.033)	0.115*** (0.030)	4.822** (1.927)
District-cohort FE	X	X	X	X
Year-st-gr-subj-coh FE	X			
Year-st-gr-coh FE		X	X	
Year-state-coh FE				X
Enroll. shares				
Adj. R^2	0.856	0.845	0.882	0.936
N	1,047,701	507,449	540,248	85,854

Notes. Estimates and standard errors of linear combinations of the parameters β_k in equation (3). The dependent variables are pooled test scores (column (1)); math and reading/ELA test scores (columns (2) and (3), respectively); and the house price index (column (4)). All columns control for district-by-cohort and cohort-by-state-by-year fixed effects. Column (1) also controls for cohort-by-state-by-year-by-grade-by-subject fixed effects, and columns (2) and (3) control for cohort-by-state-by-year-by-grade fixed effects. Standard errors in parentheses are clustered at the district level. * $p < .1$; ** $p < .05$; *** $p < .01$.

decade, denoted as $\sum_{\ell=1}^{10} K_{jct-\ell}$:

$$\begin{aligned} Y_{jrwt} &= \alpha_{jc} + \mu_{s(j)rwt} + \rho \sum_{\ell=1, \dots, 10} K_{jct-\ell} \\ (5) \quad &+ \sum_{k \neq 0} [\psi_k M_{jt-k} + P^g(V_{jct-k}, \pi_k^g)] + u_{jrwt}. \end{aligned}$$

In this model, ρ is the per dollar effect of changes in cumulative spending per pupil on test scores. We estimate equation (5) via OLS using the predictions from the first-stage stacked DRD as explanatory variables. To account for the two-step procedure, we report bootstrapped standard errors clustered at the district level.

These estimates indicate that a \$1,000 cumulative increase in per pupil spending over 10 years increases test scores by 0.017 standard deviations (Table IV, Panel A, column (1)). To account for the long life of capital projects, we follow the literature (Neilson and Zimmerman 2014; Jackson and Mackevicius 2024,

TABLE IV
2SLS EFFECTS OF INCREASES IN CUMULATIVE CAPITAL SPENDING ON TEST SCORES AND HOUSE PRICES

Sample	Share low SES		Share minority		Capital stock	
	All (1)	Low (2)	High (3)	Low (4)	High (5)	Low (7)
Panel A: Test scores						
Cap spending (\$1,000)	0.0170** (0.008)	0.0064 (0.006)	0.0276** (0.014)	0.0182** (0.006)	0.0225** (0.012)	0.0290*** (0.011)
W/ depreciation	0.0476	0.0180	0.0773	0.0509	0.0630	0.0811
Yr-st-gr-subj-coh FE	X	X	X	X	X	X
N	1,047,701	325,126	321,312	242,084	356,114	413,106
Panel B: House prices						
Cap spending (\$1,000)	1.660*** (0.648)	0.171 (0.445)	3.271** (1.382)	0.245 (0.417)	2.932*** (1.145)	2.316** (0.961)
W/ depreciation	4.648	0.478	9.158	0.685	8.209	6.486
Yr-st-coh FE	X	X	X	X	X	X
N	94,625	40,363	20,095	24,867	30,424	31,460
Panel C: House prices						
Bond amount (\$1,000)	0.265* (0.149)	0.182 (0.126)	0.510 (0.510)	0.157 (0.099)	0.616** (0.312)	-0.022 (0.180)
W/ depreciation	0.742	0.509	1.429	0.439	1.725	-0.063
Yr-st-coh FE	X	X	X	X	X	X
N	94,625	40,363	20,095	24,867	30,424	31,460
District-cohort FE	X	X	X	X	X	X

TABLE IV
CONTINUED

Notes. 2SLS estimates and standard errors of the parameter ρ in [equation \(5\)](#). The dependent variables are standardized test scores (Panel A) and the house price index (Panels B and C). “Cap spending” denotes cumulative capital spending increases in the previous 10 years. “Bond amount” denotes cumulative amounts of authorized bond in the previous 10 years. Column (1) is estimated on the full sample of districts; columns (2) and (3) on the subsamples of districts with a share of FRPL students in the bottom and top terciles, respectively; columns (4) and (5) on the subsamples of districts with a share of minority (Black and/or Hispanic) students in the bottom and top terciles, respectively; and columns (6) and (7) on districts with a capital stock above and below the national median, respectively. In Panel A, we pool data from all grades and years and control for district-by-cohort and cohort-by-state-by-year-by-grade-by-subject fixed effects, weighing observations by the number of test takers. In Panels B and C, we control for district-by-cohort and cohort-by-state-by-year fixed effects, weighing observations by district enrollment. Coefficients in the row with depreciation are obtained considering an average life span of 30 years for capital investments and a depreciation rate of 9%. Bootstrapped standard errors in parentheses are clustered at the district level. * $p < .1$; ** $p < .05$; *** $p < .01$.

for example) and amortize the spending increase over time, assuming a project life of 30 years and a depreciation rate of 9%.²³ Under these assumptions, a \$1,000 per pupil increase in the flow value of capital spending increases test scores by 2.8 times as much, or 0.048 standard deviations.

V.C. Effects on Student Sorting and Implications for Student Achievement

A possible explanation for the test-score effects of bond authorization is that capital investments improve learning. An alternative explanation is household sorting in response to capital investments. This behavior could change the composition of a district's student body and affect districts' average achievement, even in the absence of a direct effect of spending on learning.²⁴

Although the absence of student-level data prevents us from tracking students over time, we can quantify the extent to which our test-score effects are driven by sorting by measuring changes in the demographic composition of each school district after a bond approval. [Online Appendix Figure C1](#) shows estimates of [equation \(4\)](#) obtained using, as dependent variables, a set of district observables, including the share of students who are Black, Asian, Hispanic, special-education, English learner, or entitled to FRPL; the district's poverty rate for people aged 5 to 17; rates of enrollment in private schools; and the student-teacher ratio. These estimates show some evidence of small compositional changes after an authorization, particularly a small but significant 7% decline in the share of students who are Hispanic, a 20% decline in the share of English learners, and a 4% decline in the share of FRPL students.

We assess the extent to which these compositional changes can explain our test-score effects using three tests described in detail in [Online Appendix C](#). The first aims at measuring the portion of our test-score and house price effects that can be attributed

23. [Jackson and Mackevicius \(2024\)](#) use a life span of 50 years and a depreciation rate of 4.7% for buildings, and 15 years and 16.5% for nonbuilding investments. We select an average of these values. Assuming a lower depreciation rate would yield larger effects in our context.

24. Sorting could affect test-score effects if students for whom we observe test scores after a bond authorization are different from before. It could also be a channel through which house price effects arise. If some people value certain projects, carrying them out will increase the housing demand and prices in those districts.

to changes in these characteristics ([Online Appendix C.1](#)). The second predicts the changes in test scores and house prices that we would have expected just by the change in district observables ([Online Appendix C.2](#)). The third assesses the importance of changes in districts' unobservable characteristics using the method of [Oster \(2019\)](#) to construct bounds for treatment effects that account for changes in unobservables ([Online Appendix C.3](#)); the estimated bounds are fairly tight. Taken together, these tests indicate that while household sorting following a bond authorization is nonnegligible, it likely explains at most one-third of the observed increase in test scores following a bond authorization. Effects of capital spending on test scores are important even in the absence of sorting.

V.D. Robustness

1. *Higher-Order Polynomials.* Our main stacked DRD estimates are obtained using a linear polynomial of the vote margin, with a constant slope on either side of the cutoff ($P^g(V_{jct-k}, \delta_k^g) = \delta_{1,k} V_{jct-k}$). [Online Appendix Figure A14](#) shows estimates using a linear polynomial with different slopes ($P^g(V_{jct-k}, \delta_k^g) = \delta_{L,k} V_{jct-k} + \delta_{R,k} V_{jct-k} D_{jct-k}$) and a quadratic polynomial ($P^g(V_{jct-k}, \delta_k^g) = \delta_{1,k} V_{jct-k} + \delta_{2,k} V_{jct-k}^2$). All estimates are robust to the choice of polynomials. For computational feasibility, in the rest of the article we use a linear polynomial with constant slope.

2. *Alternative Designs.* Our estimates are robust to the use of alternative designs, including the TOT estimator of CFR ([Online Appendix Figures A15 and A16](#)); our stacked DRD design where we define clean controls as districts that do not approve any bonds in the entire $[c - 5, c + 10]$ time window ([Online Appendix Figures A17 and A18](#)); our stacked DRD design where we define clean controls as districts that have the same bond history as at least one treated district (in other words, we match treated and control districts in each cohort based on their bond history; [Online Appendix Figures A19 and A20](#)); our stacked DRD design obtained without controlling for future bond histories ([Online Appendix Figures A21 and A22](#)); and the ETWFE estimator of [Wooldridge \(2021\)](#) ([Online Appendix Figures A23 and A24](#)).

VI. WHAT WORKS? DIFFERENCES IN EFFECTIVENESS ACROSS SPENDING CATEGORIES

The results presented so far indicate that the approval of school-capital bonds leads to increases in test scores. These average estimates may mask important differences in effects across districts and types of bonds. These differences could explain why some of the existing studies, using data from individual states, have found much more muted effects of capital spending on test scores. To begin unpacking this heterogeneity, we study whether bonds' effects depend on what is financed, that is, the categories of projects that each bond funds. We view this as a form of treatment heterogeneity: different projects may have disparate impacts on student learning and be valued differently by taxpayers.

VI.A. *Bond Spending Categories: Summary Statistics*

Bond categories include the expansion, renovation, and construction of classroom space (52% of all proposed bonds and 45% of approved ones); infrastructure such as plumbing, furnishing, and roofs (28% and 27%); the acquisition or upgrade of IT equipment and the furnishing of laboratories (which we denote as STEM, 27% and 28%); the purchase of transportation vehicles (24% and 31%); the construction and renovation of athletic facilities (19% and 17%); modifications to improve building safety and health standards, such as removing pollutants (18% and 20%); land purchases (14% and 13%); and the installation and replacement of HVAC systems (10% and 12%, [Table I](#) and [Online Appendix Figure A25](#), Panel A). More than two-thirds of all bonds with at least one category are assigned to more than one category, with a mean of 2.9 categories for all proposed bonds and 3.2 categories for authorized bonds, respectively ([Online Appendix Figure A25](#), Panel B, and [Online Appendix Table A1](#)). Some categories are more frequently bundled together in a single bond compared to others. For example, among all proposed bonds that include classrooms, 16.9% also include HVAC and 26.3% include athletic facilities ([Online Appendix Figure A26](#)).

VI.B. *Category-Specific Effects of Bond Authorization*

To study whether the effect of a bond authorization differs depending on the category of financed projects, we estimate the

following equation:

$$Y_{jct} = \alpha_{jc} + \gamma_{s(j)ct} + \sum_{k \neq 0} \left[\beta_{k,p} D_{jct-k,p} + \phi_{k,p} M_{jct-k,p} + P^g \left(V_{jct-k}, \delta_{k,p}^g \right) \right] + u_{jct}, \quad (6)$$

where p denotes a spending category, $D_{jct-k,p}$ equals one if district j of cohort c authorized a bond within category p in year $t - k$, and $M_{jct-k,p}$ equals one if the same district proposed a bond in category p in year $t - k$. We obtain estimates of the model parameters separately for each category by retaining, in each cohort, only treated and control districts that propose a bond within that category. Under the assumption that the returns from investing in a given category do not differ systematically across districts that propose that category and those that do not, we can interpret the parameters $\beta_{k,p}$ as causal and compare them across categories.²⁵ For simplicity, we present linear combinations of the parameters $\beta_{k,p}$ for $k \in [3, 6]$ separately for each p (we report fully dynamic estimates in [Online Appendix Figure A28](#)).²⁶ When estimating effects on test scores, we pool data for multiple grades and districts and include state-by-grade-by-subject-by-cohort-by-year fixed effects in all specifications, weighing observations by the number of test takers. Because of the presence of multiple categories in a given bond, each of these category-specific estimates correspond to weighted averages over bundles that also include that category. These estimates can be interpreted as the causal effect of a given category only under the assumption that the effect of that cate-

25. Although we can never test whether districts that propose a category and those that do not differ on the basis of unobservables, we can compare them on the basis of a rich set of observable characteristics. OLS estimates of models with an indicator for districts proposing a bond with a given category in a given year as the dependent variable and a set of district observables as the explanatory variables yield F -statistics for the joint significance of districts' observable characteristics smaller than or equal to 5 ([Online Appendix Table A3](#)). Results are similar if we estimate these specifications on our stacked data set, controlling for district-by-cohort and state-by-cohort-by-year fixed effects ([Online Appendix Table A4](#)).

26. The estimates in [Online Appendix Figure A28](#) reveal positive and significant (although small) increases in test scores in years 1–2 only for categories such as other infrastructures and STEM, which likely take little time to complete. Projects such as classrooms and athletic facilities, some of which require construction and therefore a longer completion times, yield effects 3–6 years after an authorization but not before.

gory is uncorrelated with whether the bond also contains other categories.²⁷

We find large differences in bonds' effects across categories. Authorizing bonds that fund HVAC systems produces the largest increase in test scores, equal to 0.24 standard deviations (Figure IV). This result is consistent with recent evidence on the learning losses caused by excessive heat (Park et al. 2020) and the negative productivity effects of high temperatures (for example, LoPalo 2023), and the detrimental consequences of air pollution for student achievement (Gilraine and Zheng 2024). Bonds that fund renovations of plumbing systems, roofs, furnaces, and STEM equipment also have a sizable test-score impact, equal to 0.17 standard deviations. Health and safety bonds have the third-largest impact at 0.16 standard deviations, in line with recent findings on the negative consequences of toxic materials in schools (such as lead or asbestos) on students' cognitive outcomes (Ferrie, Rolf, and Troesken 2012; Sorensen et al. 2019; Gazze, Persico, and Spirovska 2024). Bonds for the expansion of classroom space increase test scores by 0.12 standard deviations. Other categories of bonds do not produce any detectable effects on test scores. Bonds for athletic facilities have an estimated positive but statistically insignificant effect of 0.05 standard deviations. Bonds for land purchases and transportation have an effect very close to zero. Time-specific estimates indicate that the impact of HVAC bonds peaks three to five years postelection but fades out quickly, whereas the impact of bonds for other infrastructure, safety and health, and STEM fades out much more slowly. The impact of classroom bonds persists 10 years postelection (Online Appendix Figure A28, Panel A).

1. *2SLS of Capital Spending Increases by Category.* The estimation of 2SLS effects of capital spending increases on test scores by category is complicated by the fact that many bonds contain more than one category but do not specify how revenues will be split across categories. To overcome this challenge, we es-

27. For robustness, in Online Appendix Figure A27 we present estimates of equation (6) obtained controlling for indicators for other categories present in the same bond, interacted with state-by-cohort-by-year fixed effects. To the extent that the impact of other categories on outcomes is constant across bonds within each state and year, these specifications should account for any correlation between category-specific effects and bond composition. Estimates barely differ between Figure IV and Online Appendix Figure A27.

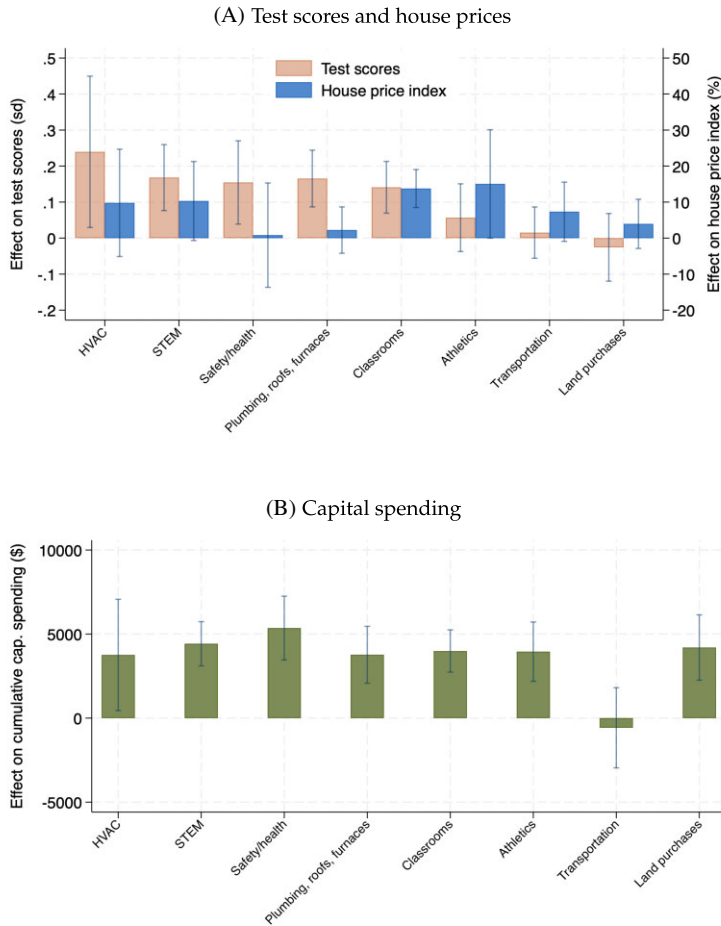


FIGURE IV

Effects of Bond Authorization by Spending Category

Point estimates and confidence intervals of an average of the parameters $\beta_{k,p}$ in equation (6) for $k \in [3, 6]$, shown separately for each spending category p . In Panel A, the orange series is estimated using test scores as the dependent variable, pooled across subjects and grades, using district-by-cohort and state-by-year-by-subject-by-grade-by-cohort effects and weighing observations by the number of test takers. The blue series is estimated using the house price index as the dependent variable, using district-by-cohort and state-by-year-by-cohort effects and weighing observations by district enrollment. In Panel B, the green series is estimated using capital spending per pupil as the dependent variable, using district-by-cohort and state-by-year-by-cohort effects and weighing observations by district enrollment. Confidence intervals are calculated using standard errors clustered at the district level.

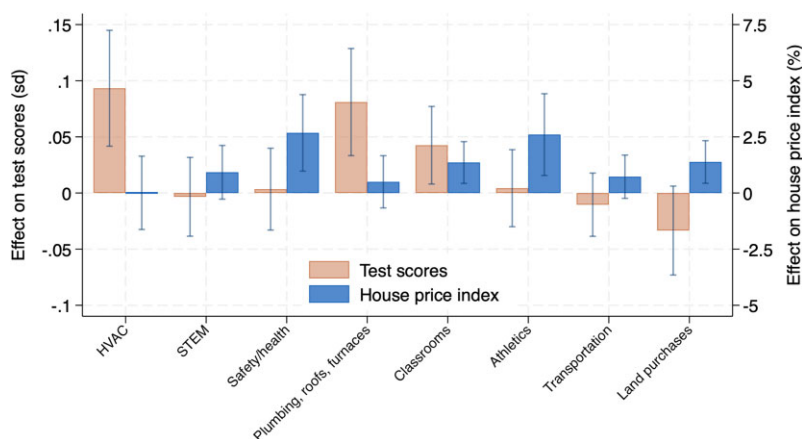


FIGURE V

2SLS Effects of Capital Spending Increases by Spending Category

2SLS point estimates and confidence intervals of the effect of spending increases on test scores and house prices, by category. Estimates are obtained by considering total spending increases (for test scores) and bond amounts (for house prices) in the first stage. The orange (light gray; color version available online) series shows effects on test scores, pooled across subjects and grades and obtained using district-by-cohort and state-by-year-by-subject-by-grade-by-cohort effects and weighing observations by the number of test takers. The blue (dark gray) series shows effects on the house price index, obtained using district-by-cohort and state-by-year-by-cohort effects and weighing observations by district enrollment. Confidence intervals are calculated using standard errors clustered at the district level.

timate models in [equation \(5\)](#) separately for each category, using the total per pupil spending increase as the endogenous explanatory variable.²⁸

2SLS estimates of the impact of spending on test scores indicate that a cumulative \$1,000 increase in per pupil capital expenditures raises test scores by 0.09 standard deviations if spent on HVAC, 0.08 standard deviations if spent on other infrastructure, and 0.04 standard deviations if spent on classrooms ([Figure V](#), orange/light gray series; color version available online). This finding largely confirms our reduced-form evidence on the test-score effects of bond authorization by category.

28. [Figure IV](#), Panel B shows increases in capital spending following the authorization of bonds in each category, obtained estimating [equation \(6\)](#) using capital spending as the dependent variable.

Taken together, although the confidence intervals of category-specific estimates often overlap, our estimates suggest that the effect of bond authorization differs depending on the spending categories. A possible explanation for this result is that districts that approve bonds in different categories serve different populations of students. For example, more disadvantaged districts could prioritize spending on HVAC systems, while more advantaged districts prioritize building stadiums. In this case, differences in effects across categories could be partly driven by differences in bond effects by student background. We explore this possibility next.

VII. WHO BENEFITS THE MOST? DIFFERENCES BY STUDENT BACKGROUND

If marginal returns to educational investments are concave, investments in school facilities may be particularly beneficial for students from more disadvantaged backgrounds. On average, these students receive much smaller private educational investments (Heckman 2008) and attend districts with lower total spending per year (Table I). Investments in school facilities can also reduce school absenteeism, which disproportionately affects students from more disadvantaged backgrounds (Lafortune and Schönholzer 2022; Baron, Hyman, and Vasquez forthcoming) and has direct implications for achievement. To investigate differences in the effect of bond authorization by student background, we test for the presence of treatment-effect heterogeneity by estimating equation (3) separately for groups of districts serving different student populations.

VII.A. *Students' Socioeconomic Status*

We begin by grouping districts according to their share of students eligible for FRPL, a proxy for low SES. We focus on districts in the bottom and top terciles of the distribution of this share across all U.S. districts in 1995. We find large differences in the impact of bond authorization across high-FRPL and low-FRPL districts. Though test scores are on a flat trend in both groups before an election, they increase rapidly in high-FRPL districts after the election, reaching a 0.15 standard deviations higher level after eight years (Figure VI, left Panel A). In low-FRPL districts, test scores are unchanged eight years postelection.

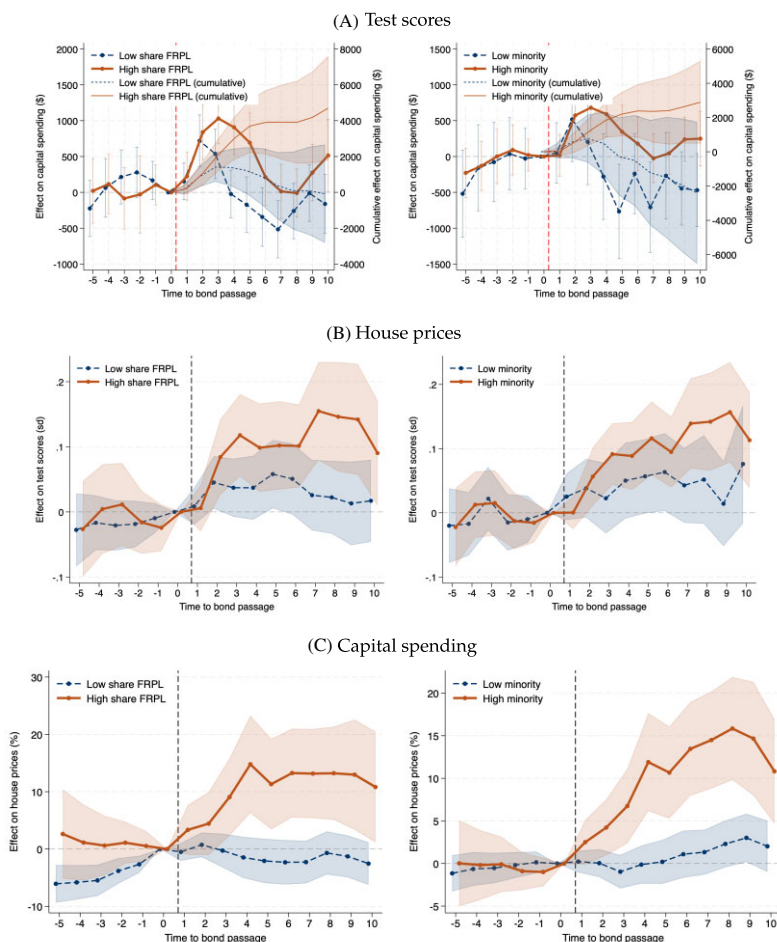


FIGURE VI

Effects of Bond Authorization by Student Demographics

Estimates and confidence intervals of the parameters β_k in equation (3), obtained using test scores (Panel A), the house price index (Panel B), and capital spending per pupil (Panel C) as the dependent variable. Figures in the left panels show estimates by tercile of the share of FRPL students (“high share” denotes the top tercile and “low share” denotes the bottom tercile). Figures in the right panels show estimates by tercile of the share of minority students (“high-minority” denotes the top tercile and “low-minority” denotes the bottom tercile). Estimates on test scores are obtained by pooling data across subjects and grades, using district-by-cohort and state-by-year-by-subject-by-grade-by-cohort effects and weighing observations by the number of test takers. Other estimates are obtained using district-by-cohort and state-by-year-by-cohort effects and weighing observations by district enrollment. Standard errors are clustered at the district level.

There are at least three reasons bond effects may be larger in high-FRPL districts. First, these districts may propose and authorize larger bonds, leading to larger increases in spending. Second, they could prioritize spending categories that increase test scores. Third, students in these districts may have higher returns on school capital investments, even conditioning on the size of the spending increase and the spending categories. Our data confirm that high-FRPL districts spend more on capital projects after a bond authorization, but these differences are not enough to explain the heterogeneity in bond impacts. 2SLS estimates indicate that a \$1,000 per pupil increase in cumulative spending increases test scores by 0.08 standard deviations in high-FRPL districts, whereas it does not produce any detectable changes in low-FRPL districts (Table IV, Panel A, columns (3) and (2), respectively). High-FRPL districts are also slightly more likely than low-FRPL ones to approve bonds to fund projects that raise test scores, such as HVAC systems, safety and health, and STEM. They are also more likely to invest in classroom space, which raises test scores (Online Appendix Figure A37).

Differences in bond composition alone are unable to explain the differences in effects between high- and low-FRPL districts. High-FRPL districts see larger effects of bond authorization within each bond category (Online Appendix Figure A35).²⁹ For example, authorizing a bond to fund HVAC systems increases test scores by 0.35 standard deviations in high-FRPL districts but leaves them unaffected in low-FRPL districts. Bonds that fund other infrastructure and STEM equipment also have larger effects in high-FRPL districts (equal to 0.38 standard deviations and 0.32 standard deviations, respectively) compared to low-FRPL districts (−0.05 and 0.01, both indistinguishable from zero). These results indicate that the larger mean effects of bond authorization on high-FRPL districts are not uniquely driven by bond composition, but also by high-FRPL students benefiting more from a given investment.

29. Online Appendix Figure A35 reproduces the estimates in Figure IV separately for high-FRPL and low-FRPL districts. We omit the transportation category due to a small number of observations.

1. *Share of Minority Students.* We also explore whether districts serving different shares of racial and ethnic minorities see different effects. We group districts in terciles of their share of Black and Hispanic (minority) students in 1995. The patterns of effect heterogeneity resemble those found in the previous section. High-minority districts (with a share of minority students in the top tercile) experience large increases in test scores after an election, equal to 0.12 standard deviations after eight years (Figure VI, right Panel A). Instead, low-minority districts (with a share in the bottom tercile) experience much smaller increases (0.06 standard deviations in test scores eight years postelection). The size of the investment and the spending categories alone cannot explain the differences in impacts: 2SLS estimates indicate that the impact of a \$1,000 increase in per pupil spending on test scores is much larger in high-minority districts (Table IV, Panel A, columns (4) and (5)). High-minority districts approve more bonds with items such as HVAC and safety and health, which increase test scores (Online Appendix Figure A37). However, they also experience larger test-score effects in those categories (Online Appendix Figure A36).

VII.B. *A Possible Mechanism: Differences in Baseline Capital Stock*

An additional possible reason for the larger bond effects in districts serving disadvantaged students is that these districts might have facilities in worse conditions. Our data indicate that high-FRPL districts tend to spend less on average. They are also more likely to be located in urban areas, which typically have older buildings (Lafortune and Schönholzer 2022). The approval of a bond may produce very different effects depending on the initial conditions of school facilities. For example, the installation or improvement of an HVAC system in a school with a deficient system may improve learning more than the replacement of the same system in a school that already had a functioning one.

We construct a measure of a district's stock of capital at any given point in time and use it as a proxy for the state of school facilities. We use historical expenditure data from the Census of Governments for 1967–2017, which record local governments' capital spending every five years. We linearly interpolate capital spending values in intercensal years and aggregate them over

30 years using a 5% depreciation rate. A district's capital stock is negatively correlated with its share of FRPL students, suggesting that, on average, more disadvantaged students attend schools in buildings in worse conditions ([Online Appendix Figure A38](#)).

To quantify differences in bond authorization effects across districts with high and low capital stock and to understand how they interact with student SES, we reestimate [equation \(3\)](#) separately for districts (i) with capital stock above and below the median in the year prior to the election, and (ii) with a share of FRPL students in the top and bottom tercile. The results of this exercise are shown in [Figure VII](#). Among districts with an above-median capital stock, the effects of bond authorization on test scores are positive but noisy. They are also only slightly larger in high-FRPL districts compared with low-FRPL ones ([Figure VII, Panel B](#)). Among districts with a below-median capital stock, instead, test-score effects are much larger in high-FRPL districts, with a 0.2 standard deviation increase seven years after an election ([Figure VII, Panel B](#)).

Taken together, our results indicate that an *ex ante* low capital stock and worse facility conditions may be responsible for some of the observed differences in bonds' impacts between high- and low-FRPL districts. Yet these differences persist even conditioning on capital stock. Therefore, students with different backgrounds may benefit more from attending schools in better conditions, particularly if these schools are not in a great state to begin with. Our findings also highlight how the detrimental effects of low levels of investments on low-SES students may compound over time, exacerbating achievement gaps between students with a different socioeconomic background.

VIII. EFFICIENCY OF SCHOOL CAPITAL INVESTMENTS

Beyond test scores, previous studies of the effects of school capital spending have examined effects on house prices, for two purposes. First, changes in house prices capture any benefits of school capital investments for students and communities not captured by test scores. If these benefits are valued by homeowners and home buyers more than the taxes they pay to finance them, spending increases should raise house prices ([Cellini, Ferreira, and Rothstein 2010](#)). Second, house prices provide a test for the efficiency of public spending. Simple models of optimal spending postulate that public goods provision is efficient when its aggre-

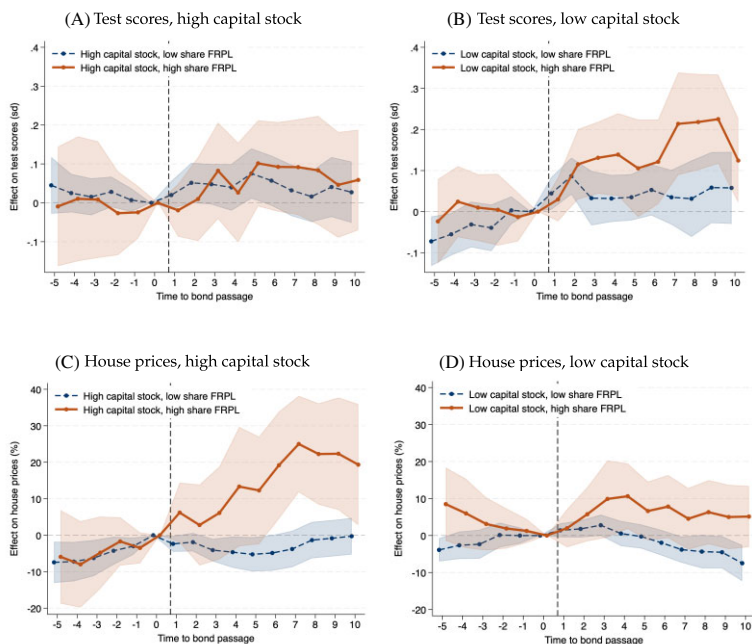


FIGURE VII

Effects of Bond Authorization by Socioeconomic Status and Initial Capital Stock

Estimates and confidence intervals of the parameters β_k in equation (3), obtained using test scores (Panels A and B) and the house price index (Panels C and D), shown separately by capital stock and share of FRPL students. “Low capital stock” and “high capital stock” denote districts in the bottom and top 50% of the distribution of the capital stock the year before the election, respectively. “High share FRPL” and “low share FRPL” denote districts in the top and bottom terciles of the distribution of the share of FRPL students. Capital stock is calculated using data from the Census of Governments for 1967–2017 as the sum of capital spending over a period of 30 years, to which we apply a depreciation rate of 5%. Estimates on test scores are obtained by pooling data across subjects and grades, using district-by-cohort and state-by-year-by-cohort-by-subject-by-grade effects and weighing observations by the number of test takers. Effects on house prices are obtained using district-by-cohort and state-by-year-by-cohort effects and weighing observations by district enrollment. Standard errors are clustered at the district level.

gate marginal benefit equals the marginal cost of providing it (Samuelson 1954). When the amount is inefficient (too high or too low), households will “vote with their feet” and re-sort across communities (Tiebout 1956). Existing residents may also be willing to pay a greater share of their income on housing if the marginal benefits are higher than their (tax) cost. Brueckner (1979) com-

bined these insights to suggest a simple test of efficiency of public good provision: if a spending increase raises house prices, the initial spending level was inefficiently low. Vice versa, if the spending increase lowers house prices, the initial level was too high.

VIII.A. Mean Effects on House Prices

To estimate the impact of increased school capital spending on house prices, we use a district-level house price index as the outcome variable in [equation \(3\)](#). This variable is normalized to 100 for each district in 1989; effects can be interpreted in percent over the mean for that year. Estimates of β_k are indistinguishable from zero before the election, indicating similar pre-election differences among districts that approve a measure in some year and those that reject it. After the election, house prices increase in districts that approve a bond measure, reaching a 9% higher level eight years postelection ([Figure III](#), Panel B). On average, house prices are 7% higher 5–8 years after a bond authorization and 5% higher 9–12 years later ([Table III](#), column (4)). This finding indicates that on average, households value increases in school capital spending more than the additional local taxes they are asked to contribute.³⁰

1. *2SLS.* [Table IV](#), Panel B shows 2SLS estimates of the effect of cumulative spending increases on house prices. We obtain these by predicting cumulative spending using, as the first stage, the model in [equation \(3\)](#) and then reestimating [equation \(5\)](#) using house prices as the dependent variable (we control for district-by-cohort and state-by-year-by-cohort). These estimates confirm that a \$1,000 per pupil increase in spending over 10 years raises house prices by 4.7% accounting for depreciation ([Table IV](#), Panel B, column (1)).

A possible interpretation for this finding is that the level of spending on school facilities is on average inefficiently low. However, neither the reduced-form estimates nor the 2SLS estimates account for the fact that several states provide districts with grants to partially cover capital expenditures. Because these grants are not financed exclusively via local taxes, they raise the marginal benefit of spending without raising marginal costs for households in the district.

30. These house price estimates are robust to the use of alternative designs ([Online Appendix Figures A16, A18, A20, A22, and A24.](#))

To conduct a proper test of spending efficiency in the presence of state grants, we estimate house price effects of increases in local capital spending by substituting cumulative lagged spending in [equation \(5\)](#) with cumulative lagged local spending (defined as the per pupil bond size proposed by a district over the past decade) and using the house price index as the dependent variable. These estimates indicate that a \$1,000 per pupil increase in local spending produces only a small increase in house prices, equal to 0.7% accounting for depreciation ([Table IV](#), Panel C, column (1)). We can thus conclude that on average, school capital spending is efficient across the United States. However, we show below that inefficiencies exist in some contexts.

VIII.B. House Price Effects and Efficiency of Spending by Category

We obtain effects of bond authorization on house prices by reestimating [equation \(6\)](#) using the HPI as the dependent variable. We find that the categories that most increase test scores are not the same as those that most increase house prices. House prices increase following the construction of athletic facilities (a 15% increase, [Figure IV](#)), the expansion of classroom space (14%), and STEM improvements (10% standard deviation). However, they remain unchanged following the approval of bonds in other test score-enhancing categories, including HVAC and safety/health. The correlation between test-score and house price estimates is -0.07 . This result suggests that home buyers value different kinds of improvements in school facilities—particularly those that are visible and have an amenity value, such as athletic facilities—over those that improve student learning. Estimates are largely robust to the use of alternative estimation procedures ([Online Appendix Figure A29](#)).

1. *Efficiency by Category: 2SLS.* To properly gauge spending efficiency by category, the blue (dark gray) series in [Figure V](#) shows 2SLS effects of bond amounts on house prices. A \$1,000 per pupil increase in (locally financed) capital spending increases house prices by 2.5 percentage points when allocated to safety and health or athletics, and 1.5 percentage points when allocated to land purchases or classrooms. These estimates indicate that spending on these categories tends to be inefficiently low. Es-

timates for other categories are instead indistinguishable from zero, indicating efficient spending levels.

VIII.C. Efficiency by Student Background

We test for differences in the house price effect of bond authorization across districts serving students from different backgrounds. We find large differences in the impact of bond authorization on house prices across districts with high and low shares of FRPL districts. In high-FRPL districts, house prices increase by 13% eight years after a successful bond election (Figure VI, left Panel B). In low-FRPL districts, instead, the effect of bond authorization on house prices is indistinguishable from zero. Differences in bond size or state aid are not enough to explain these differences: 2SLS estimates of bond amounts on house prices are also larger in high-FRPL districts (1.4% compared with 0.51% in low-FRPL districts), despite being imprecisely estimated (Table IV, Panel C, columns (3) and (2), respectively). This finding suggests that school capital investments are inefficiently low in high-FRPL districts. We obtain similar results when grouping districts according to their share of minority students (Figure VI, right Panel B, Online Appendix Figure A37, and Table IV, columns (4) and (5)).

VIII.D. Efficiency by Baseline Capital Stock

Last, we explore whether differences in baseline facility conditions may explain some of the heterogeneity in the house price effects. As before, we reestimate equation (3) separately for districts (i) with capital stock above and below the median in the year prior to the election, and (ii) with a share of FRPL-eligible students in the top and bottom tercile. Among districts with an above-median capital stock, house prices increase by more than 10% four years after an election in high-FRPL districts, whereas they do not change (and, if anything, they decline slightly) in low-FRPL districts (Figure VII, Panel C). Among districts with a below-median stock, the impact is larger for high-FRPL districts, and equal to 12% and 21% four and eight years after an election, respectively (Figure VII, Panel D). Overall, these results indicate that an ex ante low capital stock is unlikely to explain the large house price effects of bond authorization on low-SES districts.

IX. DISCUSSION AND CONCLUSION

We investigate the effect of investments in school capital on student learning and the real estate market, studying what types of investments work and under which circumstances. Using variation from closely decided school-bond referenda, we show that the approval of a bond increases test scores by 0.1 standard deviations and house prices by 7% five to eight years after an election in the average U.S. district. These estimates indicate that investing in school facilities is beneficial for students and valued by the community. Using 2SLS, we show that the increase in house prices is primarily due to the presence of state aid, rather than to inefficiencies in the *ex ante* level of spending.

These average effects mask significant variation across funded projects and across districts serving different student populations. Investments in school infrastructure such as HVAC, safety and health, and plumbing, roofs, and furnaces produce large increases in test scores but no effects on house prices, possibly because they are not “visible” to homeowners without school-age children. School investments that carry an amenity component, such as the construction or renovation of athletic facilities, raise house prices even if they do not affect learning. Districts that serve more socioeconomically disadvantaged students tend to pass more bonds with larger effects on test scores and house prices. In part due to their focus on these types of investments (although not entirely), low-SES and minority students see the largest benefits from bond authorization.

IX.A. Using Our Estimates to Better Target Spending

Our estimates can inform policies to improve overall achievement and close achievement gaps across districts serving different student populations. To illustrate this point, we quantify how much of the achievement gaps between districts serving different shares of FRPL students could be closed simply by (i) equalizing spending across districts, raising it to the level of the highest spending districts and (ii) targeting funds toward projects with the largest achievement effects.

Raising capital spending per pupil in low-SES districts to the level of high-SES districts would imply an increase in spending of about \$1,000 per pupil over 10 years in low-SES districts

(Table I).³¹ Absent a change in the distribution of spending across categories, this spending increase would raise test scores by about 0.08 standard deviations in low-SES districts (using the estimate in Table IV, Panel A, column (3)). The same increase would also close about 8% of the initial achievement gap between high- and low-SES districts, equal to roughly one standard deviation (Table I). However, increasing spending by the same amount and targeting it toward categories that are most effective in raising test scores, such as HVAC or other infrastructure, has the potential of generating increases in test scores roughly three times as large. A proper targeting of resources could therefore close up to 25% of the achievement gap between high- and low-SES districts.

IX.B. Reconciling the Estimates from the Literature

The differences in effects across bonds and districts can rationalize the contrasting results found in previous state-level studies. Although our state-level estimates differ slightly from previous studies (likely due to different outcome variables, data aggregation, and empirical models), in Online Appendix Figure A39, Panel A we replicate the positive effects found in Ohio by Conlin and Thompson (2017), the small positive effects found in California (Cellini, Ferreira, and Rothstein 2010; Rauscher 2020), and the lack of effects in Wisconsin (Baron 2022), Michigan (Conlin and Thompson 2017), and Texas (Martorell, Stange, and McFarlin 2016). Our data further indicate that differing effects across states and districts appear to be driven by the composition of the student body and the spending categories. In Online Appendix Table A5 we decompose the variance of district-specific test-score effects into various dimensions of heterogeneity using a Shapley-Owen decomposition (Israeli 2007; Huettner et al. 2012) and show that most of the total variation in test-score effects can be explained by demographic characteristics and bond categories combined. These results underscore the importance of accounting for differences in these factors when assessing the impact of capital spending on students.

31. On average, spending on capital per pupil differs by \$97 per year between low- and high-SES districts. Hence, closing this difference for 10 years implies a cumulative spending increase of about \$970 per pupil.

SUPPLEMENTARY MATERIAL

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

DATA AVAILABILITY

The data underlying this article are available in the Harvard Dataverse, <https://doi.org/10.7910/DVN/HCRJYJ> (Biasi, Lafortune, and Schönholzer 2025).

YALE UNIVERSITY SCHOOL OF MANAGEMENT, UNITED STATES
PUBLIC POLICY INSTITUTE OF CALIFORNIA, UNITED STATES
UNIVERSITY OF CALIFORNIA, SANTA CRUZ, UNITED STATES

REFERENCES

- Abott, Carolyn, Vladimir Kogan, Stéphane Lavertu, and Zachary Peskowitz, "School District Operational Spending and Student Outcomes: Evidence from Tax Elections in Seven States," *Journal of Public Economics*, 183 (2020), 104142. <https://doi.org/10.1016/j.jpubeco.2020.104142>.
- Baron, E. Jason, "School Spending and Student Outcomes: Evidence from Revenue Limit Elections in Wisconsin," *American Economic Journal: Economic Policy*, 14 (2022), 1–39. <https://doi.org/10.1257/pol.20200226>.
- Baron, E. Jason, Joshua Hyman, and Brittany Vasquez, "Public School Funding, School Quality, and Adult Crime," *Review of Economics and Statistics*, forthcoming. https://doi.org/10.1162/rest_a_01452.
- Biasi, Barbara, "School Finance Equalization Increases Intergenerational Mobility," *Journal of Labor Economics*, 41 (2023), 1–38. <https://doi.org/10.1086/718980>.
- Biasi, Barbara, Julien Lafortune, and David Schönholzer, "Replication Data for: 'What Works and for Whom? Effectiveness and Efficiency of School Capital Investments Across the United States'," Harvard Dataverse, 2025. <https://doi.org/10.7910/DVN/HCRJYJ>.
- , "School Capital Expenditure Rules and Distribution," *AEA Papers and Proceedings*, 111 (2021), 450–454. <https://doi.org/10.1257/pandp.20211040>.
- Black, Sandra E., and Stephen Machin, "Housing Valuations of School Performance," in *Handbook of the Economics of Education*, vol. 3, Eric A. Hanushek, Stephen Machin, and Ludger Woessman, eds. (Amsterdam: North-Holland, 2011), 485–519. <https://doi.org/10.1016/B978-0-444-53429-3.00010-7>.
- Blagg, Kristin, Fanny Terrones, and Victoria Nelson, "Assessing the National Landscape of Capital Expenditures for Public School Districts," Research Report, Urban Institute, Washington, DC, 2023.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess, "Revisiting Event-Study Designs: Robust and Efficient Estimation," *Review of Economic Studies*, 91 (2024), 3253–3285. <https://doi.org/10.1093/restud/rdae007>.
- Brueckner, Jan K., "Property Values, Local Public Expenditure and Economic Efficiency," *Journal of Public Economics*, 11 (1979), 223–245. [https://doi.org/10.1016/0047-2727\(79\)90006-9](https://doi.org/10.1016/0047-2727(79)90006-9).
- Brunner, Eric, Ben Hoen, and Joshua Hyman, "School District Revenue Shocks, Resource Allocations, and Student Achievement: Evidence from the Universe of U.S. Wind Energy Installations," *Journal of Public Economics*, 206 (2022), 104586. <https://doi.org/10.1016/j.jpubeco.2021.104586>.

- Callaway, Brantly, and Pedro H. C. Sant'Anna, "Difference-in-Differences with Multiple Time Periods," *Journal of Econometrics*, 225 (2021), 200–230. <https://doi.org/10.1016/j.jeconom.2020.12.001>.
- Candelaria, Christopher A., and Kenneth A. Shores, "Court-Ordered Finance Reforms in the Adequacy Era: Heterogeneous Causal Effects and Sensitivity," *Education Finance and Policy*, 14 (2019), 31–60. https://doi.org/10.1162/edfp_a_00236.
- Card, David, and Alan B. Krueger, "Does School Quality Matter? Returns to Education and the Characteristics of Public Schools in the United States," *Journal of Political Economy*, 100 (1992a), 1–40. <https://doi.org/10.1086/261805>.
- , "School Quality and Black-White Relative Earnings: A Direct Assessment," *Quarterly Journal of Economics*, 107 (1992b), 151–200. <https://doi.org/10.2307/2118326>.
- Cellini, Stephanie Riegg, Fernando Ferreira, and Jesse Rothstein, "The Value of School Facility Investments: Evidence from a Dynamic Regression Discontinuity Design," *Quarterly Journal of Economics*, 125 (2010), 215–261. <https://doi.org/10.1162/qjec.2010.125.1.215>.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer, "The Effect of Minimum Wages on Low-Wage Jobs," *Quarterly Journal of Economics*, 134 (2019), 1405–1454. <https://doi.org/10.1093/qje/qjz014>.
- Coleman, James S., Ernest Campbell, Carol Hobson, James McPartland, Alexander Mood, Frederick Weinfeld, and Robert York, *Equality of Educational Opportunity*, (Washington, DC: National Center for Educational Statistics, 1966).
- Conlin, Michael, and Paul N. Thompson, "Impacts of New School Facility Construction: An Analysis of a State-Financed Capital Subsidy Program in Ohio," *Economics of Education Review*, 59 (2017), 13–28. <https://doi.org/10.1016/j.econedurev.2017.05.002>.
- Contat, Justin, and William D. Larson, "A Flexible Method of Housing Price Index Construction Using Repeat-Sales Aggregates," *Real Estate Economics*, 52 (2024), 1551–1583. <https://doi.org/10.1111/1540-6229.12474>.
- Cornman, S. Q., O. Ampadu, K. Hanak, M. Howell, and S. Wheeler, "Revenues and Expenditures for Public Elementary and Secondary School Districts: FY 19. Finance Tables. NCES 2021-304," National Center for Education Statistics, Washington, DC, 2021.
- Deshpande, Manasi, and Yue Li, "Who Is Screened Out? Application Costs and the Targeting of Disability Programs," *American Economic Journal: Economic Policy*, 11 (2019), 213–248. <https://doi.org/10.1257/pol.20180076>.
- Dube, Arindrajit, Daniele Girardi, Òscar Jordà, and Alan M. Taylor, "A Local Projections Approach to Difference-in-Differences," Working Paper no. 31184, National Bureau of Economic Research, Cambridge, MA, 2023. <https://doi.org/10.3386/w31184>.
- Enami, Ali, James Alm, and Rodrigo Aranda, "Labor versus Capital in the Provision of Public Services: Estimating the Marginal Products of Inputs in the Production of Student Outcomes," *Economics of Education Review*, 83 (2021), 102131. <https://doi.org/10.1016/j.econedurev.2021.102131>.
- Fahle, Erin M., Belen Chavez, Demetra Kalogrides, Benjamin R. Shear, Sean F. Reardon, and Andrew D. Ho, "Stanford Education Data Archive Technical Documentation version 4.1," 2021.
- Ferrie, Joseph P., Karen Rolf, and Werner Troesken, "Cognitive Disparities, Lead Plumbing, and Water Chemistry: Prior Exposure to Water-Borne Lead and Intelligence Test Scores among World War Two U.S. Army Enlistees," *Economics & Human Biology*, 10 (2012), 98–111. <https://doi.org/10.1016/j.ehb.2011.09.003>.
- Filardo, Mary, "State of Our Schools: America's K–12 Facilities 2016," 21st Century School Fund, Washington, DC, 2016.
- Gazze, Ludovica, Claudia Persico, and Sandra Spirovska, "The Long-Run Spillover Effects of Pollution: How Exposure to Lead Affects Everyone in

- the Classroom,” *Journal of Labor Economics*, 42 (2024), 357–394. <https://doi.org/10.1086/723175>.
- Gilraine, Michael, and Angela Zheng, “JUE insight: Air Pollution and Student Performance in the U.S.,” *Journal of Urban Economics*, 143 (2024), 103686. <https://doi.org/10.1016/j.jue.2024.103686>.
- Goncalves, Felipe, “The Effects of School Construction on Student and District Outcomes: Evidence from a State-Funded Program in Ohio”, available at SSRN 2686828, 2015. <https://doi.org/10.2139/ssrn.2686828>.
- Goodman-Bacon, Andrew, “Difference-in-Differences with Variation in Treatment Timing”, *Journal of Econometrics*, 225 (2021), 254–277. <https://doi.org/10.1016/j.jeconom.2021.03.014>
- Handel, Danielle Victoria, and Eric A. Hanushek, “US School Finance: Resources and Outcomes,” in *Handbook of the Economics of Education*, vol. 7, Eric A. Hanushek, Stephen Machin, and Ludger Woessmann, eds. (Amsterdam: North-Holland, 2023), 143–226. <https://doi.org/10.1016/bs.hesedu.2023.03.003>
- Hanushek, Eric A., “Assessing the Effects of School Resources on Student Performance: An Update,” *Educational Evaluation and Policy Analysis*, 19 (1997), 141–164. <https://doi.org/10.3102/01623737019002141>.
- Heckman, James J., “The Case for Investing in Disadvantaged Young Children,” DICE Report, CESifo, Munich, 2008.
- Hong, Kai, and Ron Zimmer, “Does Investing in School Capital Infrastructure Improve Student Achievement?” *Economics of Education Review*, 53 (2016), 143–158. <https://doi.org/10.1016/j.econedurev.2016.05.007>.
- Huettner, Frank, and Marco Sunder, “REGO: Stata Module for Decomposing Goodness of Fit According to Owen and Shapley Values,” in *United Kingdom Stata Users’ Group Meetings 2012*, vol. 17, (Stata Users Group, 2012).
- Hyman, Joshua, “Does Money Matter in the Long Run? Effects of School Spending on Educational Attainment,” *American Economic Journal: Economic Policy*, 9 (2017), 256–280. <https://doi.org/10.1257/pol.20150249>.
- Israeli, Osnat, “A Shapley-Based Decomposition of the R-Square of a Linear Regression,” *Journal of Economic Inequality*, 5 (2007), 199–212. <https://doi.org/10.1007/s10888-006-9036-6>.
- Jackson, C. Kirabo, “Does School Spending Matter? The New Literature on an Old Question,” in *Confronting Inequality: How Policies and Practices Shape Children’s Opportunities*, R. Dunifon, L. Tach, and D. L. Miller, eds. (Washington, DC: American Psychological Association, 2020), 165–186. <https://doi.org/10.1037/0000187-008>.
- Jackson, C. Kirabo, and Claire L. Mackevicius, “What Impacts Can We Expect from School Spending Policy? Evidence from Evaluations in the United States,” *American Economic Journal: Applied Economics*, 16 (2024), 412–446. <https://doi.org/10.1257/app.20220279>
- Jackson, C. Kirabo, Rucker C. Johnson, and Claudia Persico, “The Effects of School Spending on Educational and Economic Outcomes: Evidence from School Finance Reforms,” *Quarterly Journal of Economics*, 131 (2016), 157–218. <https://doi.org/10.1093/qje/qjv036>.
- Kraft, Matthew A., “Interpreting Effect Sizes of Education Interventions,” *Educational Researcher*, 49 (2020), 241–253. <https://doi.org/10.3102/0013189X20912798>.
- Lafortune, Julien, and David Schönholzer, “The Impact of School Facility Investments on Students and Homeowners: Evidence from Los Angeles,” *American Economic Journal: Applied Economics*, 14 (2022), 254–289. <https://doi.org/10.1257/app.20200467>.
- Lafortune, Julien, Jesse Rothstein, and Diane Whitmore Schanzenbach, “School Finance Reform and the Distribution of Student Achievement,” *American Economic Journal: Applied Economics*, 10 (2018), 1–26. <https://doi.org/10.1257/app.20160567>.

- LoPalo, Melissa, "Temperature, Worker Productivity, and Adaptation: Evidence from Survey Data Production," *American Economic Journal: Applied Economics*, 15 (2023), 192–229. <https://doi.org/10.1257/app.20200547>.
- Martorell, Paco, Kevin Stange, and Isaac McFarlin, "Investing in Schools: Capital Spending, Facility Conditions, and Student Achievement," *Journal of Public Economics*, 140 (2016), 13–29. <https://doi.org/10.1016/j.jpubeco.2016.05.002>.
- McCrary, Justin, "Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test," *Journal of Econometrics*, 142 (2008), 698–714. <https://doi.org/10.1016/j.jeconom.2007.05.005>.
- Neilson, Christopher A., and Seth D. Zimmerman, "The Effect of School Construction on Test Scores, School Enrollment, and Home Prices," *Journal of Public Economics*, 120 (2014), 18–31. <https://doi.org/10.1016/j.jpubeco.2014.08.002>.
- Oster, Emily, "Unobservable Selection and Coefficient Stability: Theory and Evidence," *Journal of Business & Economic Statistics*, 37 (2019), 187–204. <https://doi.org/10.1080/07350015.2016.1227711>.
- Park, R. Jisung, Joshua Goodman, Michael Hurwitz, and Jonathan Smith, "Heat and Learning," *American Economic Journal: Economic Policy*, 12 (2020), 306–339. <https://doi.org/10.1257/pol.20180612>.
- Rauscher, Emily, "Delayed Benefits: Effects of California School District Bond Elections on Achievement by Socioeconomic Status," *Sociology of Education*, 93 (2020), 110–131. <https://doi.org/10.1177/0038040719892577>.
- Reardon, Sean F., Andrew D. Ho, Benjamin R. Shear, Erin M. Fahle, Demetra Kalogrides, Heewon Jang, and Belen Chavez, "Stanford Education Data Archive (Version 4.1)," 2021. <https://purl.stanford.edu/xv742vh9296>.
- Reardon, Sean F., Benjamin R. Shear, Katherine E. Castellano, and Andrew D. Ho, "Using Heteroskedastic Ordered Probit Models to Recover Moments of Continuous Test Score Distributions From Coarsened Data," *Journal of Educational and Behavioral Statistics*, 42 (2017), 3–45. <https://doi.org/10.3102/1076998616666279>.
- Samuelson, Paul A., "The Pure Theory of Public Expenditure," *The Review of Economics and Statistics*, 36 (1954), 387–389. <https://doi.org/10.2307/1925895>.
- Sorensen, Lucy C., Ashley M. Fox, Heyjie Jung, and Erika G. Martin, "Lead Exposure and Academic Achievement: Evidence from Childhood Lead Poisoning Prevention Efforts," *Journal of Population Economics*, 32 (2019), 179–218. <https://doi.org/10.1007/s00148-018-0707-y>.
- Sun, Liyang, and Sarah Abraham, "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects," *Journal of Econometrics*, 225 (2021), 175–199. <https://doi.org/10.1016/j.jeconom.2020.09.006>.
- Tiebout, Charles M., "A Pure Theory of Local Expenditures," *Journal of Political Economy*, 64 (1956), 416–424. <https://doi.org/10.1086/257839>.
- U.S. Department of Education, Institute of Education Sciences, "Public School Revenue Sources. Condition of Education," Technical Report, Institute of Education Sciences, Washington, DC, 2023.
- Wooldridge, Jeffrey M., "Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators," SSRN working paper, 2021. <https://doi.org/10.2139/ssrn.3906345>.